Paper 3: Temporal Dynamics and Universal Remodeling — Impedance Debt, Sleep, Lifecycle, Aging, and Twenty-Nine C2 Networks

Hsi-Yu Huang[1,*]

[1] *Independent Researcher, Taipei, Taiwan*

(Dated: March 17, 2026)

Papers 1 and 2 of this series established the spatial architecture of impedance-minimising networks (relay-node emergence, fractal scaling, Murray's Law, dual neural–vascular organ health $H = (1 - \Gamma_n^2)(1 - \Gamma_v^2)$). Here we develop the full *temporal* theory in two complementary parts.

Part A introduces *impedance debt* $D_Z = \int |\Gamma|^2 P_{\text{in}}\, dt$: the time-integrated mismatch energy that accumulates during wakefulness. We show that sleep is the minimum-action mechanism for discharging D_Z—a thermodynamic necessity, not a behavioural choice. Decomposing D_Z into microscopic (synaptic drift) and thermal (waste heat) components explains why jellyfish sleep without a brain (D_{thermal} removed by ocean) and why the water-to-land transition imposes a physical constraint favouring centralised neural architecture (D_{thermal} must be managed internally). A dual-field extension (Z-field + material field ρ) unifies Wolff's law (bone), Davis's law (muscle), and impedance-remodeling (neurons) as parameter instances of a single impedance-matching PDE, with embryogenesis recapitulating the thermodynamic causal chain (heart → neural tube → brain).

Part B derives the *Lifecycle Equation*,

$$\frac{d(\Sigma\Gamma^2)}{dt} = \underbrace{-\eta\,\Sigma\Gamma^2}_{\text{learning}} + \underbrace{\gamma\,\Gamma_{\text{env}}(t)}_{\text{novelty}} + \underbrace{\delta\,D(t)}_{\text{aging}},$$

governing the complete cognitive trajectory from birth to senescence as a three-force competition on the dual neural–vascular network. The resulting *bathtub curve* of $\Sigma\Gamma^2$ reproduces the known developmental trajectory without any fitted function. Emotion, reward, stress, curiosity, and boredom emerge as *readouts* of Γ dynamics at different time scales: dopamine encodes $-\Delta\Gamma^2$, cortisol encodes chronic Γ^2 elevation, curiosity encodes the impedance gradient, and boredom encodes stagnation at non-zero mismatch. Aging is derived as Coffin–Manson fatigue accelerated by Arrhenius stress kinetics, with atherosclerosis as vascular fatigue and the cascade $\Gamma_v\uparrow \rightarrow \rho\downarrow \rightarrow \Gamma_n\uparrow$ explaining why cardiovascular disease precedes neurodegeneration.

Part C proves that C2 is the *universal* remodeling law: we map all twenty-nine known tissue-remodeling mechanisms to the single rule $\Delta Z = -\eta\,\Gamma\,x_{\text{in}}\,x_{\text{out}}$, identify the thermodynamic boundaries of life (ice, fire, Goldilocks 37°C), and derive the two central controllers (cardiac pump, regulatory brain) whose failure propagates Γ error to every downstream organ. Together, Parts A–C establish that temporal dynamics, lifecycle, and tissue adaptation are all consequences of $\mathcal{A}[\Gamma] \rightarrow \min$.

Part A: Impedance Debt and Sleep

I. INTRODUCTION

Paper 1 [1] showed that the Minimum Reflection Action $\mathcal{A}[\Gamma] = \int |\Gamma|^2\, dt$ on an impedance-heterogeneous network forces hierarchical relay-node emergence, with dimensional cut cost A_{cut} irreducible by any local rule; fractal scaling recovers Kleiber's $B \sim M^{3/4}$ and Murray's branching exponent. Paper 2 [2] extended the framework to vascular networks and established the dual-network organ model $H = (1 - \Gamma_n^2)(1 - \Gamma_v^2)$.

These results are *spatial*: they describe the equilibrium architecture of impedance networks. But biological networks operate in time. Action potentials flow, ion channels drift, metabolic heat accumulates. What is the cost of running the network?

This paper extends the Minimum Reflection Action to the time domain. Part A defines impedance debt D_Z—the integral of $|\Gamma|^2$ weighted by incident power—and shows that sleep is its minimum-action discharge mechanism. The decomposition of D_Z into microscopic and thermal components provides physical constraints consistent with cnidarian sleep, the water-to-land neural transition, and embryogenesis as a single thermodynamic causal chain. Part B derives the Lifecycle Equation—the first-order ODE governing $\Sigma\Gamma^2$ from birth to death—and shows that cognition, emotion, and aging are three forces competing on the same impedance network.

* llc.y.huangll@gmail.com

II. PHYSICAL FRAMEWORK

A. Impedance mismatch and the three constraints

At every signalling interface, the reflection coefficient is

$$\Gamma = \frac{Z_{\text{load}} - Z_{\text{source}}}{Z_{\text{load}} + Z_{\text{source}}}\,. \tag{1}$$

The three inviolable constraints are:

- **C1 (Energy Conservation)**: $\Gamma^2 + T = 1$ at every interface, every tick.
- **C2 (Impedance Remodeling)**: $\Delta Z = -\eta\,\Gamma\,x_{\text{in}}\,x_{\text{out}}$.
- **C3 (Impedance-Tagged Transport)**: All inter-module signals are `ElectricalSignal` objects carrying Z-metadata.

B. Neurons as transmission lines

A myelinated axon has characteristic impedance

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}\,, \tag{2}$$

where R, L, G, C are the distributed per-unit-length resistance, inductance, conductance, and capacitance of the neural membrane. At the synapse, Z_0 meets the postsynaptic input impedance Z_{post}, producing a reflection coefficient $\Gamma = (Z_{\text{post}} - Z_0)/(Z_{\text{post}} + Z_0)$. Perfect transmission ($\Gamma = 0$) requires impedance matching; any mismatch wastes energy as reflected power.

Over time, ion-channel conductances drift (fatigue, temperature fluctuation, molecular turnover), shifting both Z_0 and Z_{post} away from their initial values. This drift accumulates as *impedance debt*.

C. Definition of impedance debt

Define the impedance debt at time t as

$$\boxed{D_Z(t) = \int_0^t \frac{1}{N}\sum_{i=1}^{N} |\Gamma_i(t')|^2\, P_{\text{in},i}(t')\,dt'} \tag{3}$$

where N is the number of signalling interfaces and $P_{\text{in},i}$ is the incident power at interface i. D_Z has units of energy and is monotonically non-decreasing during wakefulness (active signalling guarantees $P_{\text{in}} > 0$).

When the impedance landscape changes slowly relative to firing rate, $\langle|\Gamma|^2\rangle$ is approximately constant over a wake epoch. Then D_Z grows linearly:

$$D_Z(t) \approx \langle|\Gamma|^2\rangle\,\bar{P}_{\text{in}}\,t\,. \tag{4}$$

Proposition 1 (Impedance debt monotonicity)**.** *During wakefulness ($P_{\text{in}} > 0$), impedance debt $D_Z(t)$ is monotonically non-decreasing. Sleep ($P_{\text{in}} \to 0$) is the only state in which D_Z can decrease.*

Proof. By definition (Eq. 3), $dD_Z/dt = N^{-1}\sum_i |\Gamma_i|^2 P_{\text{in},i} \geq 0$ whenever $P_{\text{in},i} > 0$. Equality holds only when all $P_{\text{in},i} = 0$ (global decoupled state = sleep). □

D. Adenosine as a D_Z readout

Adenosine accumulates extracellularly in proportion to metabolic demand—specifically, to ATP hydrolysis rate, which is proportional to $P_{\text{dissipated}} = \Gamma^2 \cdot P_{\text{in}}$. Thus extracellular adenosine concentration $[\text{Ade}](t) \propto D_Z(t)$. This identifies adenosine as a *local impedance-debt meter*—consistent with its established role as the primary molecular correlate of sleep pressure (Borbély's Process S [10]).

E. D_Z decomposition: microscopic and thermal

The total impedance debt decomposes into two physically distinct components:

$$D_Z = D_{\text{micro}} + D_{\text{thermal}}\,. \tag{5}$$

D_{micro} (synaptic impedance drift) is the cumulative mismatch from ion-channel stochastic drift, receptor desensitisation, and molecular turnover. D_{thermal} (waste heat) is the Γ^2-derived dissipation that raises local tissue temperature. This decomposition has physical consequences: organisms immersed in high-κ media (ocean) can passively dissipate D_{thermal} but must still manage D_{micro} via sleep-like quiescence.

F. Dual-network impedance debt

Paper 2 [2] established that organ health depends on two impedance networks. Impedance debt likewise splits:

$$D_Z = D_{Z,n} + D_{Z,v}\,, \tag{6}$$

where $D_{Z,n}$ is neural and $D_{Z,v}$ is vascular debt. The discharge time constants differ by orders of magnitude: neural debt discharges on the timescale of a night's sleep ($\eta_n \sim 0.08$); vascular debt discharges on weeks to months ($\eta_v \sim 0.001$). This asymmetry means nightly sleep primarily services $D_{Z,n}$, while $D_{Z,v}$ accumulates chronically unless offset by vascular remodeling. The coupling from Paper 2, $I_{\text{blood}} = (1 - \Gamma_v^2)\,Q_0$, means rising vascular debt ($\Gamma_v\uparrow$) starves neural repair of material ($\rho\downarrow$), amplifying neural debt in a positive-feedback loop.

G. Sleep as impedance-debt discharge

Sleep—formally, a *global decoupled dissipation state* in which external drive is removed ($P_{\rm in} \to 0$) so that the intrinsic restorative term $-\eta\,\Gamma^2$ dominates the impedance ODE—serves as the system's offline maintenance phase. During NREM sleep, the brain activates a recalibration mode: reduced sensory input halts D_Z accumulation, while slow oscillations drive Z-map restoration via C2 impedance remodeling. The discharge rate follows:

$$D_Z(t) = D_Z^{\rm peak}\, e^{-\beta_{\rm recal}\,(t-t_{\rm sleep})}\,, \tag{7}$$

where $\beta_{\rm recal}$ is the N3 recalibration rate. The organism sleeps until $D_Z < D_Z^*$ (threshold).

The maximum tolerable wake duration before D_Z exceeds a critical threshold D_Z^* is:

$$\boxed{T_{\rm wake}^{\rm max} = \frac{D_Z^*}{\langle|\Gamma|^2\rangle\,\bar{P}_{\rm in}}} \tag{8}$$

This explains why infants ($\langle\Gamma^2\rangle \approx 0.12$) sleep $\sim$16 h/day while adults ($\langle\Gamma^2\rangle \approx 0.04$) need only $\sim$8 h: lower average mismatch means slower debt accumulation.

a. Quantitative link to glymphatic clearance. The exponential D_Z discharge (Eq. 7) has a direct biophysical correlate: the glymphatic system. Xie et al. showed that interstitial space expands by $\sim$60% during NREM sleep, accelerating cerebrospinal-fluid (CSF) convective clearance of solutes including amyloid-β [37]. Fultz et al. subsequently demonstrated that large-amplitude CSF oscillations are phase-locked to NREM slow waves [38]. In Γ-Net language, glymphatic flow is the *physical transport mechanism* that realises D_Z discharge: reduced sensory drive ($P_{\rm in} \to 0$) allows C2 remodeling to dominate, while the 60% expansion of interstitial space lowers the effective hydraulic impedance $Z_{\rm ISF}$, increasing the transmission coefficient $T = 1 - \Gamma^2$ for waste-clearance channels. The prediction is quantitative: $\beta_{\rm recal}$ in Eq. 7 should scale with CSF flow rate, and any intervention that reduces glymphatic efficiency (e.g. aquaporin-4 knockout [39]) should prolong the time required for $D_Z < D_Z^*$, i.e. increase sleep need—consistent with observation.

H. Forced awakening and the gate-leak inequality

Sleep requires that all thalamic gates satisfy $G_i \approx 0$, so the television inequality (derived in the companion Paper 4 [4]) falls below threshold: $Z_{\rm arousal}\sum_i G_i{\cdot}Q_{\rm PFC} < \theta$. However, a gate is an impedance attenuator, not a perfect open circuit. Even at $G_i \ll 1$ the *leaked power* on channel i is

$$P_{{\rm leak},i} \;=\; G_i^2\,\Gamma_i^2\,P_{{\rm in},i}\,. \tag{9}$$

The organism is forced awake whenever *any single channel's leak exceeds the Johnson–Nyquist noise floor*:

$$\boxed{G_i^2\,\Gamma_i^2\,P_{{\rm in},i} \;>\; 4\,k_B T\,R_{\rm eff}\,\Delta f \;\equiv\; P_\theta} \tag{10}$$

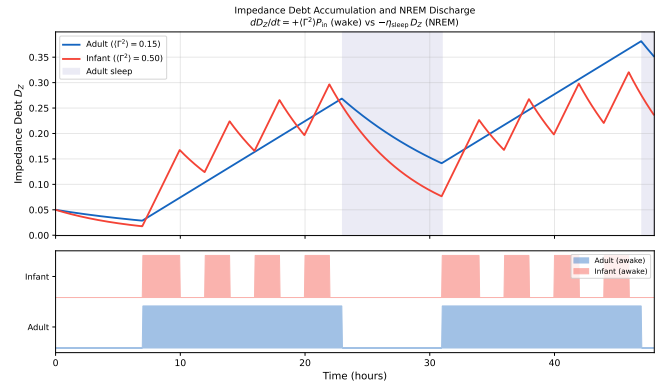

FIG. 1. Impedance debt D_Z accumulation and NREM discharge over 48 hours. Adult (blue, $\langle\Gamma^2\rangle = 0.15$, monophasic): D_Z rises during wake and drops exponentially during NREM. Infant (red, $\langle\Gamma^2\rangle = 0.50$, polyphasic): higher Γ^2 drives faster accumulation, necessitating frequent naps. Bottom: wake/sleep state timeline for both profiles.

This is the **gate-leak inequality**: the physical condition for forced awakening. Three routes satisfy it:

1. **External threat**: $P_{{\rm in},i}$ spikes (loud noise, pain, predator vibration). The startle circuit (see Paper 4 [4] for full derivation) drives $G_{\rm threat} \to 1$ within $\sim$5 ms, far faster than cortical processing. This is the ancestral survival mechanism.

2. **Endogenous alarm**: Γ_i^2 is chronically elevated on an internal channel (anxiety, rumination, pain). Even with G_i nominally low and $P_{{\rm in},i}$ at resting baseline, the product $\Gamma_i^2 P_{{\rm in},i}$ exceeds the gate's attenuation capacity. The channel "bleeds through" and re-satisfies the television inequality $\to$ **insomnia**.

3. **Thermal discomfort**: $T_{\rm body}$ drifts outside the Arrhenius operating range ($T < T_c$ or fever $T > T_{\rm max}$), hypothalamic thermoreceptors fire at high $P_{\rm in}$, and the thermal channel leaks through $\to$ awakening.

Insomnia (route 2) is therefore not a "failure to relax" but a *C1 consequence*: a channel whose Γ_i^2 is structurally forged at high mismatch (trauma, chronic stress) radiates enough leaked power to keep the consciousness loop above threshold. The gate is doing its job—it simply cannot attenuate a signal that strong. Treatment must reduce Γ_i^2 itself (C2 remodelling over long time horizons—Paper 3, Remark 4) or reduce $P_{{\rm in},i}$ (removing the stressor), not "try harder to sleep": gating G_i is already at its physical minimum.

I. Narcolepsy and hypersomnia: the arousal-collapse dual

Insomnia is the gate-leak failure mode: too much signal penetrates a closed gate. Narcolepsy and hypersomnia are the *dual* failure: the arousal or gating system itself is structurally compromised.

a. Narcolepsy (arousal anchor loss). The television inequality (see Paper 4 [4]) requires $Z_{\text{arousal}} > Z_{\min}$ for consciousness. Z_{arousal} is held at its operating point by the orexin (hypocretin) system, which acts as a *structural impedance anchor*:

$$Z_{\text{arousal}}(t) \;=\; Z_{\text{anchor}} + \delta Z(t)\,, \qquad |\delta Z| \ll Z_{\text{anchor}}\,. \tag{11}$$

When the orexin-producing neurons are destroyed (autoimmune ablation in type-1 narcolepsy), $Z_{\text{anchor}} \to 0$ and Z_{arousal} becomes a random walk:

$$Z_{\text{anchor}} = 0 \;\implies\; P(Z_{\text{arousal}} < Z_{\min}) \;\propto\; t^{1/2}\,. \tag{12}$$

The system crosses threshold at unpredictable intervals → **sleep attacks**. Cataplexy (sudden muscle-tone loss) is the motor-gate analogue: an emotional Γ^2 spike in the limbic system leaks through the brainstem motor-gate and drives $G_{\text{motor}} \to 0$—the same gate-leak mechanism as Eq. (10), but acting on the descending motor channel rather than the ascending sensory channel.

b. Hypersomnia (β_{recal} deficit). In idiopathic hypersomnia (IH), the arousal anchor is intact ($Z_{\text{anchor}} > 0$) and there is no gate-leak. Instead, the NREM recalibration rate β_{recal} in Eq. (7) is structurally low:

$$\beta_{\text{recal}}^{\text{IH}} \;\ll\; \beta_{\text{recal}}^{\text{normal}} \;\implies\; T_{\text{sleep}}^{\text{IH}} = \frac{1}{\beta_{\text{recal}}^{\text{IH}}} \ln \frac{D_Z^{\text{peak}}}{D_Z^{*}} \;\gg\; T_{\text{sleep}}^{\text{normal}}\,. \tag{13}$$

The organism requires far longer sleep to discharge the same D_Z, and waking before $D_Z < D_Z^*$ produces "sleep inertia"—the persistent fog of unfinished recalibration. Excessive daytime sleepiness follows because residual D_Z raises $\langle\Gamma^2\rangle$ on all channels, so the gate-leak condition $G_i^2\,\Gamma_i^2\,P_{\text{in}} < P_\theta$ *barely holds*: the smallest increase in P_{in} (e.g. a warm room, monotonous stimuli) tips any channel past threshold and the system collapses back into sleep.

c. Diagnostic table.

Disorder	Broken parameter	Physics
Insomnia	Γ_i^2 too high	Leak $> P_\theta$ while gates closed Cannot close
Narcolepsy	$Z_{\text{anchor}} \to 0$	Arousal random walk
Hypersomnia	β_{recal} too low	D_Z never fully discharged

All three are *structural* impedance failures (different Z_s parameters) sharing a common C1 framework. None is a "psychological" disorder; each specifies which physical parameter must be targeted for treatment: lower Γ_i^2 (insomnia), restore Z_{anchor} (narcolepsy, via orexin agonists), or raise β_{recal} (hypersomnia, via GABA_A modulation or wake-promoting agents that reduce residual D_Z).

J. NREM/REM partition

NREM sleep discharges D_Z (impedance recalibration). REM sleep reorganises the Z-map topology (adding/pruning connections). The required REM fraction scales with topological complexity:

$$T_{\text{REM}} \propto N_{\text{topo}}\,, \tag{14}$$

where N_{topo} is the number of active nodes in the impedance network. This explains the high REM fraction in neonates (∼50%, rapidly growing N) versus adults (∼25%, stable N).

III. DUAL-FIELD EXTENSION: MORPHOGENESIS

When impedance varies in space (tissue structure) and is coupled to a material-supply field, the framework becomes a reaction–diffusion system:

$$\frac{\partial Z}{\partial t} = D_Z\,\nabla^2 Z \underbrace{-\eta\, f(\rho)\,\Gamma\, Z}_{\text{C2 remodeling}} \underbrace{- E(\Gamma^2)\,(-\Gamma)\,\rho}_{\text{enzyme}} - \lambda\, Z\,, \tag{15}$$

$$\frac{\partial \rho}{\partial t} = D_\rho\,\nabla^2 \rho + I_{\text{blood}} - c(\Gamma^2)\,\rho\,, \tag{16}$$

where:

- D_Z, D_ρ are diffusion coefficients,
- $f(\rho) = \rho/(\rho + K_\rho)$: material-limited repair (Michaelis–Menten form),
- $E(\Gamma^2)$: Hill enzyme kinetics,
- $I_{\text{blood}} = (1-\Gamma_v^2)\,Q_0$: vascular supply from Paper 2,
- λ: entropic degradation rate.

The factor $(-\Gamma)$ in the enzyme term ensures bidirectional remodeling: building when $Z < Z_0$ ($\Gamma < 0$, osteoblast-like) and degrading when $Z > Z_0$ ($\Gamma > 0$, osteoclast-like)—the mathematical expression of Wolff's law [11]. Under appropriate parameters, the same PDE reduces to Davis's law (muscle) and impedance remodeling (neurons).

Gate-leak status

When the diffusion ratio D_ρ/D_Z exceeds a tissue-specific threshold, the uniform steady state becomes unstable via Turing instability [12], producing spatial patterns in the Z-field (bone trabecular architecture, hepatic lobule periodicity).

Gates collapse suddenly

Gates have Z-field holding

The material-bottleneck limit ($\rho \to 0$) yields:

$$\rho \to 0 \;\implies\; \frac{\partial Z}{\partial t} \to -\lambda\, Z\,. \tag{17}$$

Only entropic degradation survives—no repair is possible without material supply.

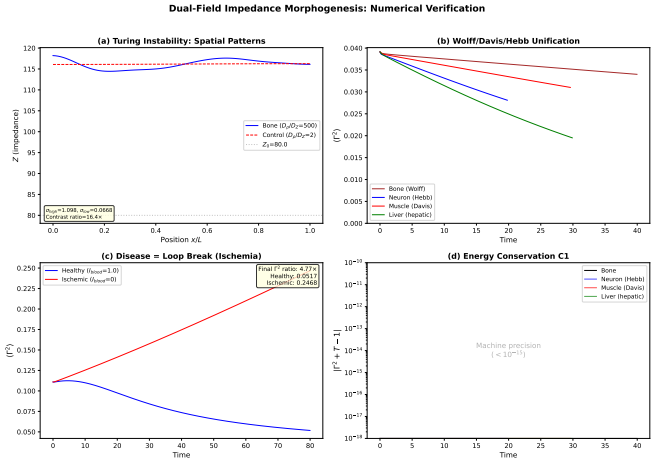

FIG. 2. Dual-field morphogenesis simulation. Spatial impedance field $Z(x,t)$ and material-supply field $\rho(x,t)$ evolve under Eqs. (15)–(16). Turing instability produces periodic spatial patterns when D_ρ/D_Z exceeds a tissue-specific threshold. Simulated via `exp_morphogenesis_pde.py`.

IV. EVIDENCE I: CNIDARIAN SLEEP WITHOUT A BRAIN

Cassiopea satisfies all sleep criteria (reduced activity, increased arousal threshold, homeostatic rebound [6]). Appelbaum *et al.* [5] showed that neuronal DNA double-strand breaks accumulate linearly with wake duration and are repaired during sleep.

In the impedance-debt framework, DNA damage is a *downstream marker* of D_Z: the causal chain is $\Gamma^2 \uparrow\rightarrow T\downarrow\rightarrow$ mitochondrial compensation $\rightarrow$ ROS $\rightarrow$ DNA damage. Jellyfish, immersed in an ocean with volumetric heat capacity $\sim$3 500$\times$ that of air, passively dissipate D_{thermal}. Yet D_{micro} (ion-channel drift) still accumulates, requiring sleep-like quiescence for discharge. Hence: **no brain is needed, but sleep is still required.**

V. PHYSICAL CONSTRAINTS ON NEURAL ARCHITECTURE

Complex nervous systems appear independently in at least nine metazoan lineages [7]. The Γ-framework does not explain this convergence but identifies the *physical constraints* that any nervous system must satisfy.

A. Impedance constraint on active foraging

An organism that transitions from passive filter-feeding to active predation gains a new signalling channel with lower Γ^2, but this channel requires coordinated impedance management across sensors, appendages, and a controller (i.e., a nervous system). Each additional sensory–motor loop increases the irreducible cut cost A_{cut} (Paper 1). The Γ-framework thus provides a *necessary condition*: any organism with N coordinated channels must pay at least $A_{\text{cut}}(N)$ in reflected energy. Whether evolution actually follows this path is determined by natural selection, not by Γ alone.

B. Thermal constraint on the water-to-land transition

Organisms on land exchange seawater ($\kappa \geq 1\ \mathrm{W\,m^{-1}K^{-1}}$) for air ($\kappa \leq 0.04$). D_{thermal} can no longer be passively removed. The Γ-framework identifies three *physical necessities* for any land-dwelling organism with significant neural complexity: (i) a convective heat-transport network (vasculature—Paper 2), (ii) a centralised impedance-debt manager (brain), (iii) structured sleep architecture for D_{micro} discharge (NREM/REM). These are constraints, not causal explanations; natural selection determines *which* lineages actually colonise land. Cephalopods remained marine but attained high metabolic rates that challenge oceanic heat-sink capacity, consistent with their distributed brain architecture (nine ganglia)—Experiment A7 at $\kappa_{\text{cross}} \approx 0.30$.

The tidal zone acts as a quarter-wave impedance matcher: $Z_{\text{match}} = \sqrt{Z_w Z_l}$, with a logarithmic κ gradient reducing the transition action $\mathcal{A}[\Gamma]$ to 6.2% of a direct jump (Experiment A6). The framework thus predicts that tidal zones offer the lowest-action path between aquatic and terrestrial impedance regimes—a testable geometric prediction, not an evolutionary cause.

C. Thermal refugia: the D_Z-clearance boundary crisis

Once on land, high-K organisms face a second constraint. To discharge accumulated impedance debt D_Z, the system must enter a low-power state (sleep), reducing metabolic heat generation to a basal level P_{basal}. System temperature during sleep obeys

$$T_{\text{sys}} = T_{\text{env}} + P_{\text{basal}}\, R_{\text{th}}\,, \tag{18}$$

where R_{th} is the organism's effective thermal resistance—a single lumped parameter integrating fur, blubber, posture, and conspecific huddling. The C2 boundary condition demands $T_{\text{sys}} > T_c(K)$ at all times; violation halts remodelling and freezes the impedance mismatch in place.

During nocturnal or seasonal temperature drops, T_{env} may swing by tens of kelvin. When $P_{\text{basal}}\, R_{\text{th}}$ alone cannot guarantee the inequality, the organism must couple to an *environmental thermal low-pass filter*—any high-heat-capacity mass whose thermal inertia damps the external oscillation. At depth $z \gg \delta = \sqrt{2\kappa/(\rho c_p\,\omega)}$, the local temperature converges to the annual mean $\overline{T}_{\text{annual}}$. Ecologically such environments (subterranean caverns, tree

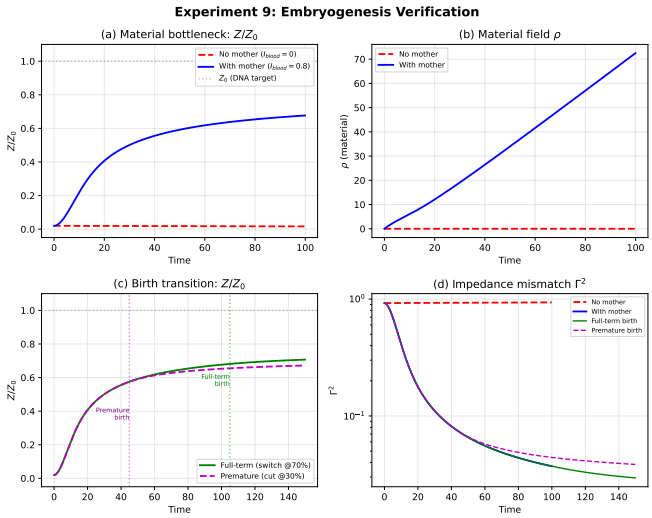

FIG. 3. Embryogenesis timeline: impedance-debt accumulation across developmental stages. Heart (day 22), neural tube (week 3–4), and brain vesicles (week 5–6) follow a thermodynamically mandatory sequence—each stage is a prerequisite for the next. Premature birth scenario (dashed) shows elevated $\Sigma\Gamma^2$ due to incomplete impedance matching. Simulated via `generate_embryogenesis_figure.py`.

hollows, snow caves, conspecific huddles) are collectively termed *thermal refugia*; physically, they are low-pass filters that replace the volatile T_{env} with a stable baseline.

The resulting constraint chain is: high $K \to$ high $T_c(K) \to$ sleep reduces $P_{\text{met}} \to P_{\text{basal}}$ R_{th} may be insufficient $\to$ external thermal coupling required. Low-K organisms ($T_c \ll 0\,°\text{C}$) satisfy the inequality with biological insulation alone. 93 tests verify the full derivation from Arrhenius $\eta(T)$ through thermal-refugium damping to survival margin.

VI. EVIDENCE III: EMBRYOGENESIS

Human embryonic development follows the thermodynamic causal chain:

1. **Heart (day 22–23)**: establishes the vascular impedance network ($I_{\text{blood}} = (1 - \Gamma_v^2)Q_0$).
2. **Neural tube (week 3–4)**: once vascular supply provides $\rho > 0$, high-metabolic neural tissue can form.
3. **Brain vesicles (week 5–6)**: centralised impedance-debt manager for both networks.

Each step is a thermodynamic prerequisite for the next. The fetus floats in amniotic fluid at ∼37.6°C, with ∼85% of metabolic heat removed via the placenta — thermodynamically a jellyfish whose heat sink is the mother. **Birth is the water-to-land transition replayed in ontogeny.**

Remark 1 (Recapitulation as impedance-threshold ordering)**.** *Haeckel's recapitulation law—"ontogeny recapitulates phylogeny"—has long been regarded as a descriptive observation without mechanistic explanation. The* Γ*-framework supplies one.*

Define the critical diffusion length*:*

$$\ell^* = \sqrt{\frac{2\,D_{\text{O}_2}\,C_{\text{surface}}}{\dot{m}}}\,, \tag{19}$$

where[?] D_{O_2} *is the oxygen diffusion coefficient in tissue,* C_{surface} *is the surface oxygen concentration, and* $\dot{m}$ *is the volumetric metabolic rate. When body thickness* $\ell < \ell^*$*, passive diffusion maintains* $\Gamma_v < 1$ *at all interior points; when* $\ell > \ell^*$*, interior cells reach* $\Gamma_v \to 1$ *(anoxia) and a dedicated pump (heart) becomes a thermodynamic necessity.*

The evolutionary timeline is then an ordered sequence of impedance thresholds:

1. ***Passive distribution*** *(*$\ell < \ell^*$*): sponges (Porifera, ∼600 Ma)—no neurons, no pump.*
2. ***Diffuse dual network*** *(*$\ell < \ell^*$*, but* Γ_n *emerges for coordination): cnidarians (jellyfish, ∼550 Ma)—nerve net, gastrovascular cavity, no centralised pump.*
3. ***Active circulation*** *(*$\ell > \ell^*$*): annelids (∼500 Ma)—first dedicated hearts.*
4. ***Internal thermal management*** *(*D_{thermal} *no longer passively dissipated): water-to-land transition (∼375 Ma)—demands centralised neural control (PFC).*

Embryogenesis replays this sequence not by historical accident but by physical causality: layer $k+1$ *of the impedance hierarchy requires* $\rho_k = (1 - \Gamma_k^2)\,Q_0 > 0$ *from layer* k*. The heart* must *precede the neural tube because neural C2 requires material supply from vascular transmission.*

Structural epochs. *This threshold ordering partitions evolutionary history into* structural epochs*—geological eras whose biota share a common impedance-threshold tier: (i) membrane epoch (>2 Ga; single-cell* Γ*-interfaces), (ii) diffusion epoch (∼600 Ma; multicellular,* $\ell < \ell^*$*), (iii) pump epoch (∼500 Ma;* $\ell > \ell^*$*, dedicated hearts), (iv) endothermic epoch (∼200 Ma; internal* D_{thermal} *management, PFC). Mass extinctions eliminate species* within *a tier but never regress the structural level: after the end-Cretaceous event (66 Ma), all surviving macrofauna retained active circulation. Regression is physically forbidden because at* $\ell > \ell^*$ *a pumpless body reaches* $\Gamma_v \to 1$ *(anoxia).*

Why the heart grows first. *Every extant vertebrate embryo develops in the pump epoch. Active circulation is therefore the minimum physical condition for subsequent tissue viability: the heart is not an evolutionary "choice" but the thermodynamic verdict of* $\ell > \ell^*$*. Genes encode*

the execution *of this physically mandatory sequence; they do not* choose *it.*

Falsifiable prediction. *No extant or fossil multicellular animal with body thickness $\ell > \ell^*$ (given its metabolic rate) survives without some form of active circulatory mechanism.*

Remark 2 (Natural selection as impedance-mismatch minimisation)**.** *The Lifecycle Equation (Theorem 1, below) governs individual organisms, but its structure applies equally at the population level. Two logically distinct selection mechanisms emerge.*

(A) Competitive selection within a structural tier. *Among organisms that share the same impedance threshold tier, those with lower $\Sigma\Gamma^2$ relative to the current environment have higher viability. This is the standard "survival of the fittest" expressed in impedance language: the fittest are the least mismatched.*

(B) Physical exclusion across tiers. *A fundamentally different mechanism operates at structural-epoch boundaries. When an environmental shift makes $\ell > \ell^*$ the norm (e.g. rising oxygen and body size in the Cambrian), organisms that* cannot *cross the pump threshold are not outcompeted—they are* physically excluded. *A body with $\ell > \ell^*$ and no active circulation reaches $\Gamma_v \to 1$ (anoxia) at its core: this is not a failure to compete but a thermodynamic impossibility. The distinction matters: competitive selection is probabilistic (better-matched organisms are more* likely *to survive); physical exclusion is deterministic (organisms that lack the required structural tier* cannot *exist).*

Mass extinction as $\Gamma_{\rm env} \to 1$. *A catastrophic environmental change (asteroid impact, supervolcanic eruption, ocean anoxia) is a global spike in $\gamma\,\Gamma_{\rm env}(t)$. When this term overwhelms the C2 correction capacity η, viability collapses. The five major mass extinctions correspond to five episodes where $\gamma\,\Gamma_{\rm env}$ exceeded η for the majority of extant species. Survivors are those with: (i) high η (rapid adaptors, e.g. bacteria), (ii) low ℓ (small body, low metabolic demand), or (iii) already low Γ in the new regime (deep-sea organisms).*

Post-extinction radiation. *When $\Gamma_{\rm env}$ relaxes, the vacated impedance landscape contains unoccupied niches. Survivors rapidly diversify to fill these niches (e.g. mammalian radiation after the end-Cretaceous event)—the $\gamma\,\Gamma_{\rm env} \to 0$ regime of the Lifecycle Equation. Crucially, no post-extinction radiation has ever regressed a structural tier: all surviving macrofauna after the end-Cretaceous retained active circulation.*

Corollary 1 (Recovery rate $= \eta(t)$)**.** *The clinically observed decline in recovery rate with age—whether from stress (Γ_n spike on neural channels), fatigue (D_Z accumulation), or wound healing (local $\Gamma_v \to 1$)—is a direct manifestation of the Arrhenius decline in $\eta(t)$. Recovery = C2 driving Γ back toward zero:*

$$\textit{Recovery rate} \;\propto\; \eta(t)\cdot x_{\rm in}\cdot|\Gamma_{\rm damage}|\,.$$

In youth, η is high, $x_{\rm in}$ is high (active lifestyle), and vascular Γ_v is low (good material supply ρ). All three factors decline with age, so recovery rate decays as their product*—faster than any single factor alone.*

Remark 3 (Exercise as C2 input-signal maintenance)**.** *The Lifecycle Equation predicts that exercise maintains $x_{\rm in} > 0$ on the vascular C2 channel (wall shear stress $\tau_{\rm wall} = 4\mu Q/\pi r^3$ increases with cardiac output Q), thereby keeping the learning term $-\eta\,x_{\rm in}\,\Sigma\Gamma^2$ active against the aging drift $\delta\,D(t)$.*

This is a zero-parameter, falsifiable prediction: physically active adults should have lower $\Sigma\Gamma^2$ than sedentary adults at every age tier.

We tested this prediction using NHANES 2013–2018 Physical Activity Questionnaire (PAQ) data (n = 7,393 adults across three 2-year cycles), cross-referenced with the pre-computed Γ-vectors from Section **??** *of Paper 5. Active respondents (n_{active} = 2,517; WHO guideline: $\geq$150 min/week moderate-equivalent) had significantly lower $\Sigma\Gamma^2$ than sedentary respondents (n_{sed} = 4,876) overall ($\bar{\Sigma\Gamma}^2_{active} = 0.634$ vs. $\bar{\Sigma\Gamma}^2_{sed} = 0.733$; Mann–Whitney $p = 3.7\times10^{-48}$) and in all three age tiers: 20–39 ($p = 8.6\times10^{-10}$), 40–59 ($p = 1.8\times10^{-8}$), 60+ ($p = 1.2\times10^{-9}$). The 60+ active group's mean $\Sigma\Gamma^2 = 0.727$ was lower than the 40–59 sedentary group's* 0.723*, consistent with the simulation prediction that a 70-year-old exerciser can have lower impedance mismatch than a non-exercising 50-year-old.*

Part B: Lifecycle Equation and Aging

VII. THE LIFECYCLE EQUATION

A. Total mismatch energy

Define the population-averaged squared reflection coefficient:

$$\Sigma\Gamma^2(t) \equiv \frac{1}{N}\sum_{i=1}^{N}\Gamma_i^2(t)\,, \tag{20}$$

with dual-network decomposition $\Sigma\Gamma^2 = \Sigma\Gamma_n^2 + w_v\,\Sigma\Gamma_v^2$, where $w_v = \alpha_{v\to n}$ is the vascular-to-neural coupling weight from Paper 2.

B. Three forces

Three physically distinct mechanisms alter $\Sigma\Gamma^2$:

(i) Learning (MRP descent). C2 impedance remodeling adapts $Z_{\rm load}$ toward $Z_{\rm source}$:

$$\left.\frac{d(\Sigma\Gamma^2)}{dt}\right|_{\rm learn} = -\eta(t)\,\Sigma\Gamma^2\,. \tag{21}$$

(ii) Novelty injection. New stimuli introduce mismatches at untuned interfaces:

$$\left.\frac{d(\Sigma\Gamma^2)}{dt}\right|_{\text{novel}} = \gamma(t)\,\Gamma_{\text{env}}(t)\,. \tag{22}$$

(iii) Aging (Coffin–Manson fatigue). Irreversible mechanical strain from Lorentz pinch ($\varepsilon_{\text{pinch}} = S|I|^{\alpha}$) drifts impedance permanently, with stress-accelerated kinetics:

$$\delta_{\text{eff}}(t) = \delta_0\,\exp\!\big(E_a\,\sigma_{\text{stress}}(t)\big)\,. \tag{23}$$

Combining all three:

Theorem 1 (Lifecycle Equation)**.** *The population-averaged squared reflection coefficient $\Sigma\Gamma^2(t)$ satisfies the non-autonomous ODE*

$$\boxed{\frac{d(\Sigma\Gamma^2)}{dt} = -\eta\,\Sigma\Gamma^2 + \gamma\,\Gamma_{\text{env}}(t) + \delta\,D(t)} \tag{24}$$

where η is the C2 learning rate, $\gamma\,\Gamma_{\text{env}}$ is the novelty injection term, and $\delta\,D$ is the irreversible Coffin–Manson aging drift. The coefficients are functions of developmental stage, environmental statistics, and cumulative damage.

Proof. The three terms follow from: (i) C2 gradient descent on $\mathcal{A}[\Gamma]$ (Eq. 21); (ii) new stimuli introducing mismatches at untuned interfaces (Eq. 22); (iii) Arrhenius-driven irreversible impedance drift (Eq. 23). Superposition gives the full ODE. □

C. Developmental modulation of $\eta(t)$

In Eq. (24), η has so far been treated as a context-dependent constant. Experimentally, synaptic plasticity, myelination rate, and tissue remodeling capacity all peak in early life and decline with age. We model this as:

$$\eta(t) = \eta_0\,f_{\text{dev}}(t)\,\exp\!\big(-E_a/k_B T_{\text{age}}(t)\big)\,, \tag{25}$$

where $f_{\text{dev}}(t)$ is a bell-shaped developmental envelope (high in infancy, plateau in adolescence, monotone decline thereafter) and the Arrhenius factor captures aging-induced loss of remodeling capacity.

The cumulative impedance change for a skill chain s acquired at time t_s is:

$$\Delta Z_s = \int_{t_s}^{\infty} \eta(\tau)\,\Gamma_s(\tau)\,x_{\text{in}}(\tau)\,x_{\text{out}}(\tau)\,d\tau\,. \tag{26}$$

Because $\eta(t_s)$ is largest in childhood and because early-acquired skills accumulate over a longer integration window, childhood-acquired skills (native language, early motor sequences) achieve the deepest ΔZ_s and lowest Γ_s.

Proposition 2 (Skill-decay anti-chronology)**.** *Under monotone decline of $\eta(t)$ and uniform aging damage $D(t)$, the order of functional loss inverts the order of acquisition:*

$$t_{s_1} < t_{s_2} \implies t_{\text{fail},s_1} > t_{\text{fail},s_2}\,.$$

Proof sketch. A deeper ΔZ_s corresponds to a lower Γ_s. Aging drives Γ_s back up at a rate independent of initial depth. The skill with the deepest ΔZ_s (earliest t_s, highest η) requires the most damage to cross the failure threshold Γ_{crit}. □

Figure 4 shows a numerical integration of three skills acquired at ages 3, 20, and 35 under the same $\eta(t)$ lifecycle and Arrhenius aging. The latest-acquired skill fails first; the childhood skill survives longest—consistent with the clinical observation that dementia erases adult-learned second languages before the native tongue, and that elderly pianists who "forgot" they could play can still perform when placed at the keyboard.

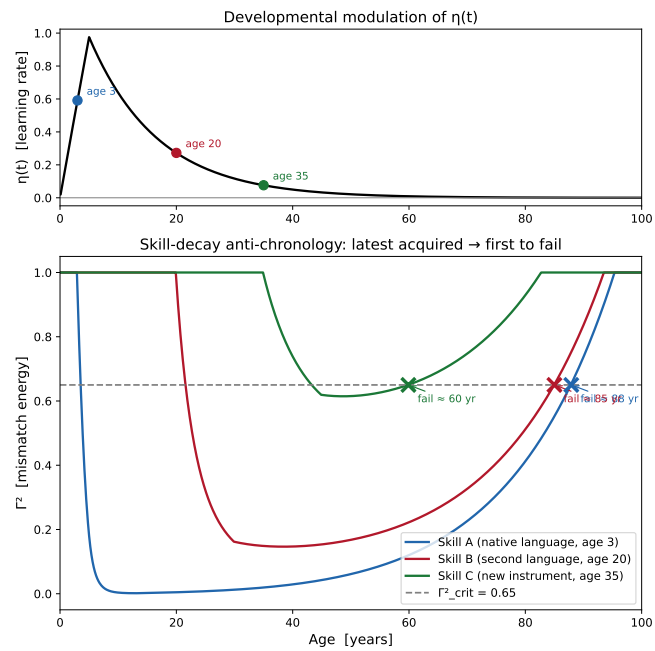

FIG. 4. Skill-decay anti-chronology. Three skills acquired at ages 3, 20, and 35 under a common $\eta(t)$ lifecycle (Eq. 25). Γ^2 is driven back up by Arrhenius aging; later-acquired skills cross the failure threshold (dashed) first.

Remark 4 (Three mechanisms of impedance fixation)**.** *The $\eta(t)$ lifecycle and C2 dynamics identify three routes by which a Z-pattern becomes functionally immutable—i.e., migrates into the effective null space $\mathcal{S}_{\text{eff}}$ (Paper 4, Remark 4):*

1. ***Critical-period forging.*** *Patterns repeatedly reinforced during the η-peak window (Eq. 25) accumulate the deepest ΔZ_s (Eq. 26). Childhood-imprinted values, fears, and attachment styles belong to this class.*

2. ***Impulsive*** Γ^2 ***spike.*** *A single event with extreme mismatch—trauma, acute humiliation, overwhelming success—produces a large instantaneous* $\Delta Z = -\eta\,\Gamma\,x_{in}\,x_{out}$ *because both* Γ *and* $x_{in}\,x_{out}$ *are momentarily very large. The resulting* Z*-shift can exceed what years of moderate stimuli would achieve, making it correspondingly harder to reverse with small corrective signals.*

3. ***Confirmatory basin trapping.*** *If the post-acquisition environment consistently reinforces the same* Z*-configuration, C2 keeps the trajectory inside the original attractor basin. By the time* $\eta(t)$ *has fallen to near-zero (Arrhenius aging), no practicable stimulus can escape the basin.*

All three routes share a common endpoint: the adult who "knows the belief is irrational but cannot change it" has correctly diagnosed that $\eta\,\Gamma\,x_{in}\,x_{out} \approx 0$ *on the relevant sub-network. This is a* physical *constraint, not a failure of will. Therapeutic change is possible but requires sustained, high-fidelity counter-stimuli over long time horizons to accumulate a measurable* ΔZ *despite the diminished* η*—analogous to cold-working a tempered steel: possible, but far slower and more limited than the original hot forging.*

D. The Equilibrium Equation

Impedance mismatch dissipates heat. Thermal balance gives the steady-state temperature:

$$T_{\text{steady}} = T_{\text{env}} + \frac{\alpha}{\beta}\,\Sigma\Gamma^2\,. \tag{27}$$

"Adulthood" has a physical definition: the thermal steady state where $d(\Sigma\Gamma^2)/dt \approx 0$ —the point at which the world can no longer burn you.

VIII. EMOTION AS IMPEDANCE DYNAMICS

Psychological states map to features of the Γ^2 time series:

A. Dopamine: the $-\Delta\Gamma^2$ signal

The reward prediction error [15] in the impedance framework is:

$$\delta_{\text{RPE}} = -\big(\Sigma\Gamma^2_{t+1} - \mathbb{E}[\Sigma\Gamma^2_{t+1}]\big)\,. \tag{28}$$

Dopamine fires when mismatch decreases *more than expected*—the impedance network matched faster than predicted.

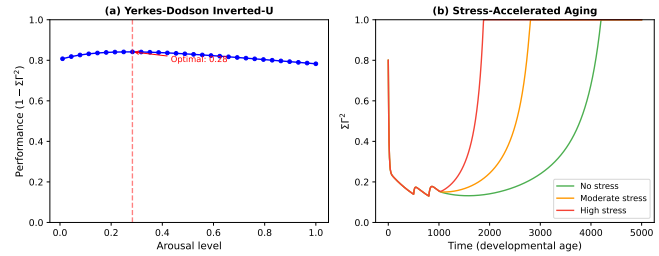

FIG. 5. Yerkes–Dodson curve derived from the Lifecycle Equation. Optimal arousal η^* depends on task difficulty ($\Sigma\Gamma^2$) and environmental novelty (Γ_{env}). The inverted-U emerges from the balance between learning rate and impedance-driven noise. Simulated via `generate_paper_iii_figures.py`.

B. Cortisol: the slow Γ^2 integrator

Stress is the time-averaged mismatch:

$$\sigma_{\text{stress}}(t) = \frac{1}{\tau_s}\int_{t-\tau_s}^{t} \Sigma\Gamma^2(t')\,dt'\,, \tag{29}$$

where $\tau_s \sim$ hours–days. Acute challenges are stimulating; prolonged elevation activates the cortisol response.

C. The Yerkes–Dodson curve

Arousal increases both η (learning) and γ (novelty intake). The optimum occurs at:

$$\eta^*\,\Sigma\Gamma^2 = \gamma^*\,\Gamma_{\text{env}} \quad\Longrightarrow\quad \text{arousal}^* \propto \frac{\Sigma\Gamma^2}{\Gamma_{\text{env}}}\,, \tag{30}$$

predicting that optimal arousal depends on task difficulty and current skill level—the Yerkes–Dodson observation [16].

D. Curiosity and boredom

Curiosity is the impedance gradient: $\mathcal{C} = -\partial(\Sigma\Gamma^2)/\partial(\text{exploration})$, gated by safety ($\Gamma^2 < \Gamma^2_{\text{threat}}$). Boredom arises when $|d\Gamma^2/dt| < \epsilon$ at non-zero mismatch—a thermodynamic imperative to seek new gradients for action reduction.

E. Metacognition

The thinking impedance $\Gamma_{\text{think}} = \sum_i w_i\Gamma_i / \sum_i w_i$ defines dual-process switching [17]: System 1 (fast, automatic) when $\Gamma_{\text{think}} < \Gamma_{\text{S2}}$; System 2 (slow, deliberate) when $\Gamma_{\text{think}} > \Gamma_{\text{S2}}$.

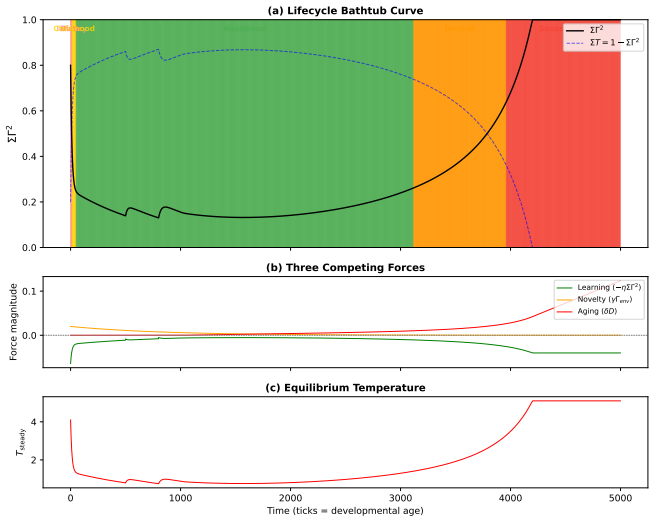

FIG. 6. Bathtub curve of cognition. Six developmental phases (birth, infancy, childhood, adulthood, decline, senescence) emerge from the Lifecycle Equation. Top panel: $\Sigma\Gamma^2(t)$; bottom panel: the three competing forces (Learning, Novelty, Aging). The trajectory mirrors the engineering reliability bathtub curve. Simulated via `generate_paper_iii_figures.py`.

IX. THE BATHTUB CURVE OF COGNITION

The Lifecycle Equation predicts six developmental phases:

1. **Birth**: $\Sigma\Gamma^2 \approx 0.8$–$1.0$. Random initialisation; Novelty $\gg$ Learning $\gg$ Aging.
2. **Infancy**: Learning dominates. η maximal (critical period); $\Sigma\Gamma^2$ drops rapidly.
3. **Childhood**: Learning $\approx$ Novelty. Intense consolidation; REM fraction high.
4. **Adulthood**: $d(\Sigma\Gamma^2)/dt \approx 0$. Equilibrium Equation gives steady-state temperature.
5. **Decline**: Aging $>$ Learning. Net mismatch increases despite continued learning.
6. **Senescence**: Aging $\gg$ Learning. Cascading failure with positive feedback.

This trajectory is formally identical to the engineering reliability *bathtub curve* [18]—not analogy but derivation from the same physics.

X. STRESS, FATIGUE, AND THE PHYSICS OF AGING

A. Coffin–Manson fatigue

Action potentials generate Lorentz forces; cumulative strain follows the Coffin–Manson law [14]: $N_f = C\,(\Delta\varepsilon_p)^{-\beta}$. "Failure" is irreversible impedance drift.

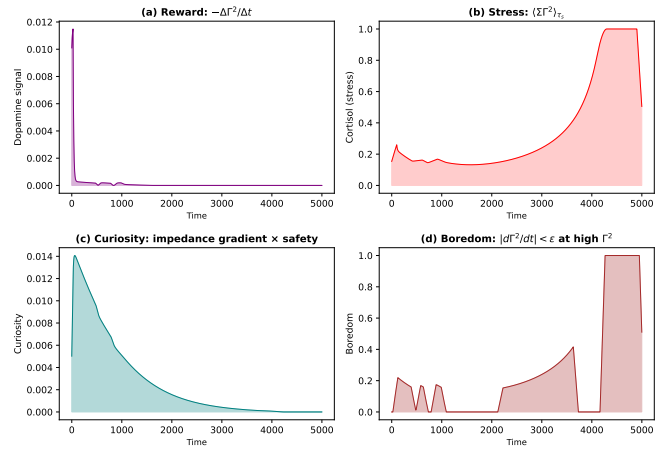

FIG. 7. Emotion readouts derived from impedance dynamics. Dopamine maps to $-\Delta\Gamma^2$ (reward prediction error), cortisol to the slow Γ^2 integrator. The four quadrants correspond to distinct emotional states (contentment, excitement, boredom, distress). Simulated via `generate_paper_iv_figures.py`.

B. Vascular Coffin–Manson: atherosclerosis

$\sim$3 billion cardiac cycles impose pulsatile strain. Cumulative vascular damage $D_v(t)$ drifts Z_v from Murray's-Law optimum (Paper 2), with plaque as the macroscopic manifestation: a 50% stenosis raises Z_v by 16$\times$ via r^{-4} Poiseuille scaling.

C. The vascular–neural aging cascade

$$D_v\uparrow \rightarrow\ \Gamma_v\uparrow \rightarrow\ \rho\downarrow \rightarrow\ \eta_{n,\mathrm{eff}}\downarrow \rightarrow\ \Gamma_n\uparrow\ . \qquad (31)$$

This explains: (i) cardiovascular disease precedes dementia by 10–20 years, (ii) blood-pressure control in midlife reduces dementia risk (SPRINT-MIND [31]), (iii) metabolic syndrome as dual-network failure.

D. Elastic versus plastic deformation

Each current pulse produces $\varepsilon = \varepsilon_{\mathrm{elastic}} + \varepsilon_{\mathrm{plastic}}$. Sleep repairs the elastic component (daily D_Z discharge); the plastic component is *irreversible*—it accumulates over a lifetime. Sleep restores daily performance but cannot prevent long-term aging [27].

E. Death as positive-feedback collapse

When $\Sigma\Gamma^2$ exceeds a critical threshold:

$$\frac{d^2(\Sigma\Gamma^2)}{dt^2} > 0 \quad\Longrightarrow\quad \text{runaway failure}\,. \qquad (32)$$

Compensatory current increases through remaining channels accelerate fatigue in a positive-feedback loop, amplified by the dual-network cascade.

F. Sealed shutdown: the alternative endpoint

Runaway collapse (Eq. 32) is not the only terminal trajectory. If, by the time life-sustaining channels approach failure, most of the individual's functional impedance has already migrated into the effective null space $\mathcal{S}_{\text{eff}}(t)$ (Sec. VII C, Paper 4 Remark 4), then:

1. the *fraction* of the network that can participate in a positive-feedback cascade is small;
2. the remaining "live" channels carry low current at low Γ, so the Coffin–Manson fatigue rate is minimal;
3. shutdown proceeds as a gentle, low-Γ^2 fade rather than a high-Γ^2 explosion.

We define the *sealing ratio*:

$$R_{\text{seal}}(t) \equiv \frac{\dim\, \mathcal{S}_{\text{eff}}(t)}{\dim\, Z_{\text{total}}} \in [0,1]\,. \tag{33}$$

A high R_{seal} at end-of-life predicts a low peak $\Sigma\Gamma^2$ during the dying process ("peaceful death"), whereas a low R_{seal} leaves many critical channels in the vulnerable high-Γ, low-η regime where positive feedback can ignite.

Clinically: individuals with deeply established life skills, stable temperament, and long sleep-health history (all proxies for high R_{seal}) should, on average, experience less terminal agitation and lower end-of-life pain burden than individuals with extensive late-acquired, poorly consolidated functional demands.

XI. SLEEP ACROSS THE LIFECYCLE

Infant sleep: $\Sigma\Gamma^2 \approx 0.8$; daily D_Z is enormous. 14–17 hours of sleep required. REM ∼50% because N_{topo} is rapidly growing.

Adult sleep: $\Sigma\Gamma^2$ near minimum; 7–9 hours suffice.

Elderly sleep: Irreversible (plastic) drift is undischargeable by sleep. Rising vascular Γ_v (atherosclerosis) reduces material supply ρ, slowing neural repair. Sleep is doubly impaired: plastic baseline rises *and* vascular-mediated repair declines [29].

XII. DEVELOPMENTAL DESIGN PRINCIPLES

A. Infant soft tissue as impedance buffer

Neonatal soft tissues (cartilage, fat, elastic skin) form *mechanical quarter-wave transformers*: $Z_{\text{infant}} \ll Z_{\text{adult}}$, attenuating impact signals before they reach nociceptors. With $\Sigma\Gamma^2 \approx 0.8$ at birth, unattenuated pain would overwhelm C2 calibration.

B. Social referencing as external Γ calibration

When an infant falls, it looks at the caregiver's face before deciding to cry [35]. The caregiver's emotional signal provides Γ_{social} superimposed on the mechanical signal: $\Gamma_{\text{perceived}} = \Gamma_{\text{mechanical}} + \eta_{\text{social}}\,\Gamma_{\text{social}}$, where $\eta_{\text{social}} \gg 1$ in infancy and $\to 0$ in adulthood (self-calibrated system). Each fall-and-look episode is a C2 calibration step: $\Delta Z_{\text{pain}} = -\eta\,\Gamma_{\text{social}}\,x_{\text{in}}\,x_{\text{out}}$.

Both strategies—structural buffering and behavioural calibration—are temporary scaffolding: bones harden and social coupling decreases as the neural network completes self-calibration.

C. Aging pain immunity

If impedance drifts upward with age, why no chronic pain? For slow drift with healthy C2 ($\eta > 0$), the impedance-remodeling update continuously re-matches: $\Gamma^2 \sim (\Delta Z/2\bar{Z})^2 \lesssim 10^{-6}$. Pain triggers only in three regimes:

1. **Acute injury** ($dZ/dt \gg \eta$): instantaneous discontinuity.
2. **C2 failure** ($\eta \to 0$): avascular tissues (cartilage) cannot remodel → osteoarthritis (Paper 5 [3]).
3. **Late senescence** ($\delta D(t) > \eta\Sigma\Gamma^2$): Coffin–Manson damage exceeds repair capacity.

XIII. SCAR TISSUE AS PERMANENT IMPEDANCE DEBT

A wound creates acute localised impedance debt: $D_{Z,\text{wound}} = \int_0^{T_{\text{repair}}} \Gamma^2_{\text{wound}}\, P_{\text{in}}\, dt$. Repair progressively reduces $|\Gamma_{\text{wound}}|$, but scar tissue never achieves $\Gamma = 0$: the residual mismatch $\Gamma^2_{\text{scar}} > 0$ is a *permanent, non-dischargeable impedance debt* analogous to the irreducible A_{cut} of Paper 1.

Clinical correlates: scar tenderness (persistent Γ^2), contracture (ongoing minimisation), keloid formation (overshoot to opposite-sign mismatch), and cumulative surgical debt (raised $\Sigma\Gamma^2$ floor).

Remark 5 (Scar debt and the bathtub curve)**.** *An individual accumulating many scars shifts baseline $\Sigma\Gamma^2$ upward, raising the bathtub-curve floor —a mechanistic link between cumulative trauma and accelerated aging.*

XIV. PTSD AND EMPATHIC LOAD AS IMPEDANCE PHYSICS

A. PTSD: anatomy of an undamped spike

The prediction "PTSD = undamped Γ^2 spike" can now be decomposed into three necessary and jointly sufficient conditions using the $\eta(t)$ lifecycle (Eq. 25) and the fixation mechanisms (Remark 4).

1. **Impulsive mismatch.** The traumatic event generates a Dirac-like Γ^2 spike—a sensory/emotional input that is "completely beyond expectation." The instantaneous C2 update $\Delta Z = -\eta\,\Gamma\,x_{\rm in}\,x_{\rm out}$ is very large because both Γ and $x_{\rm in}\,x_{\rm out}$ are momentarily extreme.

2. **High effective η at time of event.** If the event occurs during a developmental η-peak (childhood abuse) or during acute hyper-arousal (combat, sudden catastrophe), the effective learning rate is elevated, deepening the ΔZ_s well beyond what normal experience would produce. This is forced tempering: the impedance pattern is forged in a single stroke.

3. **Absent C2 resolution path.** In the aftermath, no subsequent experience provides a sustained $(x_{\rm in}, x_{\rm out})$ trajectory that could guide Γ^2 back down. The traumatic Z-pattern sits at a local minimum of $\mathcal{A}[\Gamma]$ surrounded by a high-energy basin wall; small perturbations return the system to the same attractor. As $\eta(t)$ continues to decline with age, the pattern slides toward $\mathcal{S}_{\rm eff}$—simultaneously high-Γ^2 and low-plasticity.

The two hallmark symptoms follow directly:

- *Re-experiencing / flashback*: the high-Γ^2 attractor has a wide basin of attraction; any stimulus near the basin boundary pulls the network into the full traumatic Z-configuration.

- *"I know it's irrational but I can't stop it"*: $\eta\,\Gamma\,x_{\rm in}\,x_{\rm out} \approx 0$ on the traumatic sub-network—the same physics as any deep fixation (Remark 4), but here the attractor carries pathologically high Γ^2 rather than the low Γ^2 of a well-matched skill.

B. Therapeutic physics

If PTSD is a high-Γ^2 pattern trapped in a deep basin with declining η, effective therapy must manipulate the three factors in C2:

- **Pharmacotherapy** (SSRIs, propranolol, MDMA-assisted therapy): modulates the effective η and noise floor, lowering the instantaneous $\Gamma^2 \times P_{\rm in}$ at each re-activation so that the system does not re-traumatise (i.e., does not deposit additional D_Z on each trigger). In impedance language: drugs widen the basin boundary and reduce the peak reflection per activation.

- **Exposure therapy / trauma-focused CBT**: provides repeated, safe, titrated $(x_{\rm in}, x_{\rm out})$ pairs that traverse the basin interior under controlled conditions, allowing many small, convergent C2 updates to gradually reshape the local Z-landscape. Each session contributes a small ΔZ in the corrective direction; over months, the accumulated integral (Eq. 26 in reverse) can lower the attractor's Γ^2 floor.

- **Combined treatment**: pharmacotherapy adjusts parameters (η, noise, activation threshold) to keep the system in the convergent C2 regime; psychotherapy provides the gradient. Neither alone is sufficient: drugs without new trajectories cannot reshape Z; new trajectories without parameter control risk re-traumatisation.

This is not "suppressing the memory" but physically reshaping the impedance sub-network—cold-working a tempered structure, slow and bounded, but real.

C. Empathic load and caregiver burnout

Empathy, in Γ-Net terms, is the temporary imposition of another agent's high-Γ^2 field onto one's own impedance network. When a clinician or caregiver engages with a trauma survivor, their own network must process the incoming Γ^2 pattern:

$$D_{Z,\rm care}(t) = \int_0^t \Gamma^2_{\rm other}(\tau)\; P_{\rm in,care}(\tau)\, d\tau\,, \qquad (34)$$

where $\Gamma^2_{\rm other}$ is the patient's transmitted mismatch and $P_{\rm in,care}$ is the caregiver's coupling strength (empathic sensitivity $\propto \eta_{\rm social}$).

Chronic exposure without adequate discharge (sleep, boundaries, peer support) raises the caregiver's own $\Sigma\Gamma^2$ time average (Eq. 29), activating the same cortisol– allostatic load pathway as any chronic stressor. The lifecycle equation for the caregiver becomes:

$$\frac{d(\Sigma\Gamma^2_{\rm care})}{dt} = -\eta\,\Sigma\Gamma^2_{\rm care} + \gamma\,\Gamma_{\rm env} + \delta\,D(t) + \underbrace{\lambda\,\Gamma^2_{\rm other}}_{\text{empathic load}}\,, \qquad (35)$$

where λ parameterises coupling strength. High-η individuals (high empathic sensitivity) absorb more of the patient's Γ^2 per unit time—precisely the trait that makes them effective caregivers also makes them vulnerable to burnout.

Remark 6 (Burnout is not weakness)**.** *Caregiver burnout—emotional exhaustion, depersonalisation, reduced efficacy—is the predictable endpoint of Eq.* (35) *when the empathic-load term chronically exceeds the combined learning and sleep-discharge capacity. It is an energy-balance constraint, not a character deficiency. Institutional policies that ignore $D_{Z,care}$ are, in impedance terms, running a network past its Coffin–Manson fatigue limit.*

Remark 7 (Moral blame as Γ^2-injection)**.** *The preceding analysis reveals a sharp criterion for evaluating social responses to fixation, trauma, and addiction. Any intervention—moral, institutional, or interpersonal—that addresses an individual's high-Γ^2 pattern can be classified by its effect on the C2 landscape:*

- ***Therapeutic**: provides safe (x_{in}, x_{out}) trajectories, raises effective η (pharmacologically or via trust), or lowers Γ_{env} $\Rightarrow$ $\Sigma\Gamma^2$ decreases over time.*
- ***Iatrogenic**: injects additional mismatch (shame, stigma, social exclusion) without supplying any corrective gradient $\Rightarrow$ $\Sigma\Gamma^2$ increases; the impedance debt deepens.*

Much of everyday "moralising" about trauma responses and stubborn habits falls into the second category: it adds Γ^2_{blame} on top of the original $\Gamma^2_{fixation}$, deposits additional D_Z, and offers no C2 path for repair. In the impedance framework, such blame is physically indistinguishable from secondary injury (Paper 4, Remark 4).

D. From psychological load to somatic collapse

The empathic-load equation (Eq. 35) feeds directly into the dual-network organ model of Paper 2: $H = (1-\Gamma_n^2)(1-\Gamma_v^2)$. The causal chain is:

$$\underbrace{\Sigma\Gamma^2_{\text{care}}\uparrow}_{\text{chronic stress}} \xrightarrow{\text{cortisol, Eq. (29)}} \Gamma_n\uparrow \xrightarrow{\text{autonomic}\downarrow} \Gamma_v\uparrow \xrightarrow{\rho\downarrow} \eta_{\text{eff}}\downarrow \Longrightarrow H\downarrow \tag{36}$$

1. **Neural overload.** Chronic high $\Sigma\Gamma^2$ elevates cortisol (Eq. 29), accelerating Coffin–Manson fatigue on neural channels and raising Γ_n.
2. **Autonomic spill-over.** Rising Γ_n impairs sympathetic–parasympathetic balance, drifting the brainstem setpoint Z_{setpoint} upward (allostatic load; Paper 5). This increases vascular Γ_v across all downstream organs (Paper 2, Eq. (31)).
3. **Material starvation.** Higher Γ_v reduces perfusion ρ, starving C2 repair in every tissue—including immune (affinity maturation) and metabolic (insulin, hepatic, erythropoietic) networks that Paper 5 identifies as C2 subsystems.
4. **Peripheral collapse.** Organs lowest in the perfusion hierarchy (skin, hair, nails) are the first to show visible degradation: sallow complexion, slow wound healing, brittle integument. These are not cosmetic complaints; they are the earliest macroscopic indicators of a system-wide $H\downarrow$.

The complete chain reads:

PTSD / empathic load $\rightarrow$ $\Sigma\Gamma^2\uparrow$ $\rightarrow$ Cortisol / allostatic load $\rightarrow$ (

Chronic high-Γ^2 caregiving and unresolved trauma do not stay "psychological": they accelerate dual-network aging, push immune and metabolic C2 past their repair capacity, and manifest as the familiar clinical picture of lowered immunity, metabolic syndrome, and accelerated somatic aging.

Remark 8 (Visual health perception as impedance tomography)**.** *The everyday judgment "this person looks exhausted" is, in Γ-Net terms, an unconscious* impedance tomography*. The observer's visual system integrates multiple peripheral indicators—skin perfusion (microvascular Γ_v), muscle tone (neuromuscular Γ_n), facial micro-expression latency (cortical Γ_n), wound-healing state (collagen C2 output)—and compresses them into a single scalar estimate of the subject's aggregate $H = (1-\Gamma_n^2)(1-\Gamma_v^2)$. "Looking well" corresponds to low Γ across visible channels; "looking haggard" corresponds to elevated Γ read out through the perfusion hierarchy's lowest-priority organs. The intuition is not metaphor: the observer is literally reading reflection coefficients from the subject's skin, posture, and responsiveness, and arriving at a population-calibrated prior on their dual-network health.*

E. Chronic stress as slow-forged PTSD: integral-path equivalence

The three conditions for PTSD identified in Sec. XIV A—(1) impulsive mismatch, (2) high η at the moment of trauma, (3) absent C2 resolution—define a sufficient but not unique route to structural Z_s fixation. A second route reaches the same attractor through temporal integration rather than instantaneous shock. Recall the impedance debt (Eq. 3):

$$D_Z = \int_0^T \Gamma^2(t)\, P_{\text{in}}(t)\, dt \;> D_Z^* \quad \Longrightarrow \quad Z_s \text{ locked.} \tag{37}$$

The integral is indifferent to the shape of its integrand: a Dirac-like spike and a months-long plateau yield identical D_Z if their areas match. When $\Gamma^2(t)$ is chronically elevated at sub-traumatic amplitude—interview anxiety, occupational uncertainty, prolonged social adversity—the integral crosses the irreversibility threshold D_Z^* on a timescale of months. The accumulated C2 update is

$$\Delta Z_s = \int_0^{T_{\text{chronic}}} \eta(\tau)\, \Gamma(\tau)\, x_{\text{in}}\, x_{\text{out}}\, d\tau\,, \tag{38}$$

bypassing conditions (1) and (2) while satisfying (3): there is no corrective trajectory because the stressor is *ambient and unresolvable* (e.g. awaiting a hiring decision with no feedback, sustaining living costs with no income certainty). In this regime Z_{load} is perpetually undefined—the brain cannot compute a gradient because the target impedance fluctuates between mutually exclusive outcomes—and C2 stalls while D_Z continues to grow.

The terminal state is identical to classic PTSD: Z_s locked in a high-Γ^2 attractor with $\eta\,|\,\Gamma\,x_{\text{in}}\,x_{\text{out}}| \approx 0$ on the affected sub-network. The downstream somatic cascade (Eq. 36) proceeds unchanged:

$$\Sigma\Gamma^2\uparrow \rightarrow \text{ Cortisol}\uparrow \rightarrow (\Gamma_n, \Gamma_v)\uparrow \rightarrow \text{ Immune C2 failure } \rightarrow \text{ Multi-organ disease.}$$

Two further consequences merit emphasis:

- **Hallucination from thermal noise.** As $\Gamma_i \rightarrow 1$ on sensory channels, $T_i \rightarrow 0$ (C1). Johnson–Nyquist noise ($P_\theta = 4k_BT\,R_{\text{eff}}\,\Delta f$) that would normally transmit harmlessly is instead reflected back into the processing loop and amplified—the brain interprets its own thermal noise as external signal. This is the same physics as the flashback mechanism (Sec. XIV A), but triggered by noise rather than a stimulus. Paper 4 [4] unifies this with sensory-deprivation hallucination: removing external input ($x_{\text{in}} \rightarrow 0$) causes adaptive gain divergence that likewise promotes thermal noise above the recognition threshold; both pathways converge to the same terminal state (Paper 4, Eq. (15)).
- **Immune collapse and opportunistic infection.** Via the five-channel coupling of Paper 5 (Eq. 16 therein), sustained neural $\Gamma_n^2\uparrow$ drains input power from the immune channel, raising Γ_{imm}^2 and suppressing affinity maturation ($\eta_{\text{imm}}\downarrow$). Common infections (upper respiratory, etc.) break through because immune $T_{\text{imm}} \rightarrow 0$.

Remark 9 (Anxiety spectrum as integration time)**.** *The clinical labels form a continuum parameterised by D_Z integration time: acute stress (D_Z small, minutes), generalised anxiety disorder (D_Z growing, weeks), burnout (D_Z moderate, months to years), and Complex PTSD ($D_Z > D_Z^*$, structurally locked). The impedance physics is a single equation; the diagnostic distinctions reflect how far along the integral the patient sits when clinically assessed.*

Part C: C2 as Universal Remodeling Law

XV. THERMODYNAMIC BOUNDARIES OF LIFE

Before cataloguing the twenty-nine tissue networks, we establish the thermodynamic envelope within which C2 can operate.

A. Ice: impedance freeze ($D_Z = 0$)

Below the freezing point of water, ion transport ceases ($x_{\text{in}} = x_{\text{out}} = 0$), C2 halts, and impedance debt cannot accumulate. Cryopreservation exploits this: tissue is "paused" in impedance space.

B. Fire: impedance deletion ($Z \rightarrow \infty$)

Above the protein denaturation threshold (~ 42°C sustained), molecular structure breaks down irreversibly. Impedance substrate is destroyed; $\Gamma^2 \rightarrow 1$ at all interfaces represents total reflection (death).

C. The Goldilocks temperature

At ~ 37°C: water is liquid ($x_{\text{in}}, x_{\text{out}} > 0$); thermal energy drives conformational changes ($\eta > 0$); Johnson–Nyquist noise $\sigma_Z^2 = 4k_BT\,\text{Re}(Z)\,\Delta f$ is small enough for C2 to converge yet large enough to explore the impedance landscape (annealing).

D. Post-glacial impedance landscape

Snowball Earth episodes (720–635 Ma) imposed prolonged $D_Z \approx 0$ conditions. Post-glacial warming restored thermal gradients and nutrient availability, creating an impedance landscape in which C2 could operate at high throughput. The Γ-framework identifies the *physical preconditions*: liquid water ($x_{\text{in}} > 0$), sufficient thermal energy ($\eta > 0$), and low initial impedance debt. Whether and how the Cambrian radiation exploited these conditions is a palaeontological question beyond the scope of this theory.

XVI. TWENTY-NINE Γ-NETWORKS

Table I maps all twenty-nine known remodeling laws to C2. Seventeen are distributed tissue branches; ten are organ-specific or sensory controllers; two are central controllers (Sec. XVII).

Whole-body integration. Beyond individual organ networks, all 29 blueprints are unified into a single `GammaTopology` (278 nodes, 426 edges) via dual inter-organ buses: a vascular bus (aortic trunk K=4 → arteriolar branches K=2) and a neural bus (autonomic trunk K=3 → branches K=2), with the brain_functional topology as central command hub. Disease cascades across organs emerge from the bus topology without disease-specific code (E0). The total structural action $\mathcal{A}_{\text{cut}} = 222$ is the irreducible price of multi-organ life. A thermoregulation derivation (Paper 1, Sec. 7.4) shows that blood viscosity $\tilde{\eta}(T)$ drives $Z_{\text{bus}}(T)$ via Arrhenius

kinetics; the repair time diverges doubly exponentially, $\tilde{\tau} \propto \exp[(\alpha+\beta)/T]$, establishing a critical temperature T_c that rises with brain complexity K. Endothermy is therefore a Γ-topological necessity for $K \geq 5$ brains.

Remark 10 (Material substrate vs. mathematical structure)**.** *The molecular transduction pathways implementing C2 in each tissue are well characterised in the respective literature; our contribution is the recognition that these twenty-nine mechanisms share* identical mathematical structure*—gradient descent on the reflection action—regardless of their material substrate.*

Remark 11 (Why twenty-nine C2 networks exist: the interface–relay–maintenance chain)**.** *The number twenty-nine is not arbitrary. By the Universal Relay-Node Corollary (Paper 0, Corollary* **??***), every impedance-discontinuous interface ($\Gamma \neq 0$) requires a relay structure for signal integrity. Each relay structure is itself a physical component subject to degradation ($\Gamma \to 1$ under Coffin–Manson fatigue) and therefore requires a maintenance mechanism. C2* is *that mechanism:*

$$\underbrace{\Gamma \neq 0}_{\textit{interface}} \implies \underbrace{A_{\rm cut} > 0}_{\textit{relay needed}} \implies \underbrace{\Delta Z = -\eta\,\Gamma\,x_{\rm in}x_{\rm out}}_{\textit{C2 maintenance}}$$

Thus each row of Table I exists because its corresponding interface exists. The twenty-nine C2 networks are not discovered empirically and then unified post hoc; they are predicted *by the number of impedance-discontinuous interfaces in the human body. Table II lists the interface and relay structure for each network.*

A. Unification Theorem

Theorem 2 (C2 Universality across all 29 tissues)**.** *Let $\mathcal{T} = \{1, 2, \ldots, 29\}$ be the index set of all known tissue-remodeling systems listed in Table I. For each tissue $i \in \mathcal{T}$, let $\mathcal{L}_i$ denote the empirically established tissue-specific remodeling law. Then there exists a unique physiological mapping*

$$\varphi_i : \textit{(tissue-specific physiological quantities)} \longrightarrow \big(Z_i,\ x_{in,i},\ x_{out,i}\big) \tag{39}$$

such that $\mathcal{L}_i$ is algebraically equivalent to Constraint C2 under φ_i:

$$\boxed{\mathcal{L}_i \ \equiv\ \Delta Z_i = -\eta_i\,\Gamma_i\,x_{in,i}\,x_{out,i} \quad \forall\, i \in \mathcal{T}} \tag{40}$$

where $\Gamma_i = (Z_{load,i} - Z_{source,i})/(Z_{load,i} + Z_{source,i})$. The only tissue-specific free parameters are the material constants $(\eta_i, Z_{0,i})$; the mathematical structure is uniquely determined by the First Law (Paper 0, Theorem 3).

Proof. The proof proceeds in three steps.

Step 1: Existence of φ_i. For every tissue i, the remodeling literature reports an update rule of the form $\Delta Q_i = f_i(\text{input}_i, \text{output}_i)$, where Q_i is a tissue-specific quantity (synaptic weight, bone density, vessel radius, receptor affinity, etc.). In each case, impedance is definable as the ratio of an effort variable to a flow variable (Eq. **??** of Paper 0), and the update ΔQ_i maps to ΔZ_i via a sign-preserving linear relation ($\Delta Z \propto -\Delta w$ for synapses; $\Delta Z \propto +\Delta\rho$ for bone; etc.). The mapping φ_i is therefore constructible explicitly for every row of Table I; five representative constructions are given in Section XVI B.

Step 2: Algebraic equivalence. Under φ_i, each $\mathcal{L}_i$ reduces to gradient descent on the reflection action $\mathcal{A} = \int \Gamma_i^2 P_{{\rm in},i}\,dt$ (Eq. **??** of Paper 0). By the uniqueness theorem for C2 (Paper 0, Theorem 5), gradient descent on $\mathcal{A}$ with activity gating yields $\Delta Z_i = -\eta_i\,\Gamma_i\,x_{{\rm in},i}\,x_{{\rm out},i}$ as the *unique* first-order descent direction. Hence $\mathcal{L}_i \equiv$ C2 under φ_i.

Step 3: Uniqueness of φ_i. Suppose two mappings φ_i and φ_i' both reduce $\mathcal{L}_i$ to C2. Then $\varphi_i' \circ \varphi_i^{-1}$ is an automorphism of C2 that preserves the gradient structure. The only such automorphisms are rescalings $\eta_i \to \lambda\eta_i$, $Z_{0,i} \to \mu Z_{0,i}$, which correspond to different choices of material units—not different physical content. Therefore φ_i is unique up to unit rescaling. □

Corollary 2 (Plasticity as $\eta_i > 0$)**.** *Tissue i is remodeling-capable if and only if $\eta_i > 0$. For $\eta_i \approx 0$ (cartilage, scar tissue), C2 predicts $\Delta Z_i \approx 0$ regardless of mismatch, making pathology under sustained load a thermodynamic necessity rather than a disease.*

Corollary 3 (Substrate independence)**.** *The molecular machinery implementing C2 in each tissue (NMDA receptors, RANKL/OPG, eNOS, AID hypermutation, EPO, FSH receptors, etc.) determines only the material constants $(\eta_i, Z_{0,i})$. The update equation is dictated by physics and is substrate-independent.*

B. Selected tissue proofs

We present five representative proofs; the remaining twenty-four follow identical logic with the substitutions shown in Table I.

1. Neural: Hebb's Law (1949)

Hebb's postulate [36]: when axon A repeatedly fires cell B, synaptic efficiency increases. Modern rate-based formulation: $\Delta w_{AB} = \eta\, r_A\, r_B$. Since synaptic weight w is inversely related to synaptic impedance ($Z_{\rm syn} \propto 1/w$):

$$\Delta Z_{\rm syn} = -\eta\,\Gamma_{\rm syn}\,x_{\rm in}\,x_{\rm out}\,. \quad \checkmark \tag{41}$$

STDP (spike-timing-dependent plasticity) arises from the causal structure of $x_{\rm in}$ preceding $x_{\rm out}$: LTP when pre $\to$ post ($\Gamma > 0$), LTD when reversed ($\Gamma < 0$).

TABLE I. Twenty-nine Γ-networks, each algebraically equivalent to C2: $\Delta Z = -\eta\,\Gamma\,x_{\text{in}} \cdot x_{\text{out}}$. The final entry is the C2 *null case* ($\eta \approx 0$).

#	Network	Flow	Effort	Z definition	x_{in}	x_{out}	Known law
1	Neural	Ion current	Membrane voltage	V/I (synaptic)	Pre-syn rate	Post-syn rate	Hebb's LTP
2	Skeletal	Strain rate	Force	F/v	Applied force	Bone strain	Wolff's Law
3	Vascular	Blood flow	Pressure	$\Delta P/Q$	Shear stress	Endothelial resp.	Glagov remodeling
4	Immune	Ab flux	Antigen conc.	$1/(K_a \cdot [R])$	[Ag]	$[\text{T}_{\text{FH}}]$	Affinity maturation
5	Epithelial	Strain rate	Contact force	F/v	Friction	Tissue deform.	Keratinisation
6	Muscular	Shortening vel.	Contractile force	F/v (Hill)	Neural drive	Tension	Davis's Law
7	Pulmonary	Gas flow	Airway pressure	$P_{\text{aw}}/\dot{V}$	Vent. drive	Alveolar exp.	Altitude acclim.
8	GI	Absorption flux	Nutrient conc.	$[\text{nut}]/J$	Luminal content	Epithelial uptake	Intestinal adapt.
9	Endocrine	Signalling flux	Hormone conc.	$1/(K_d \cdot N_R)$	Hormone conc.	Intracell. resp.	Receptor reg.
10	Lymphatic	Lymph flow	Interst. pressure	P/Q_{lymph}	Antigen load	Node response	Reactive hypertroph
11	Renal	GFR	Glomerular pressure	P_{glom}/Q	Macula densa	Arteriolar resp.	TG feedback
12	Hepatic	Clearance flux	Metabolite conc.	$[\text{met}]/J$	Portal signal	Prolif. resp.	Liver regeneration
13	Connective	Strain rate	Tendon force	$E \cdot A/L$	Mech. strain	Fibroblast resp.	Tendon remodeling
14	Adipose	Storage flux	Energy substrate	$[\text{sub}]/J$	Insulin signal	Adipocyte resp.	Leptin signalling
15	Haematopoietic	O_2 delivery	O_2 demand	P_{O_2}/Q_{O_2}	EPO signal	Erythroid resp.	Erythropoiesis
16	Reproductive	Maturation flux	FSH signal	$1/(N_R \cdot K_a)$	FSH conc.	Granulosa resp.	Follicular selection
17	Cardiac (ctrl)	Ejection flow	Vent. pressure	P/Q_{ej}	Wall stress	Myocardial strain	Laplace–Starling
18	Reg. brain (ctrl)	Set-point signal	Error signal	Z_{setpoint}	Sensory input	Autonomic output	Baroreceptor reset
19	Ocular	Aqueous flow	IOP	$P_{\text{IOP}}/Q_{\text{aq}}$	Aq. pressure	Trabecular drainage	IOP homeostasis
20	Auditory	Acoustic flow	Sound pressure	Cochlear Z	Sound press.	Hair cell resp.	Tonotopic plasticity
21	Vestibular	Angular flow	Accel. signal	Vestibular gain	Ang. velocity	Oculomotor resp.	VOR adaptation
22	Olfactory	Ligand flux	Odorant conc.	Receptor $1/K_d$	Odorant conc.	ORN response	Receptor turnover
23	Dental	Stress flux	Thermal/mech.	Dentin Z	Insult	Odontoblast resp.	Tert. dentinogenesis
24	Enteric	Strain rate	Luminal press.	Gut compliance	Distension	ENS firing	Enteric plasticity
25	Autonomic	Neural flux	Set-point error	Sympathovagal Z	Baroreceptor	Effector output	Sympathovagal remo
26	Pancreatic	Insulin flux	Glucose conc.	$1/(N_\beta \cdot k_{\text{sec}})$	Glucose	Insulin secr.	β-cell compensation
27	CSF	CSF flow	ICP	$P_{\text{CSF}}/Q_{\text{CSF}}$	CSF pressure	Production rate	Choroid regulation
28	Brain (func.)	Spike rate	Field potential	Synaptic Z matrix	Afferent freq.	Cortical area	Cortical map plastic
29	Cartilage ($\eta \approx 0$)	Creep	Load	$H_A \cdot A/h$	Load	Deformation	**(OA predicted)**

2. *Skeletal: Wolff's Law (1892)*

Wolff's Law [11]: bone internal architecture adapts to imposed mechanical demand. Bone impedance $Z_{\text{bone}} = F/v$ is proportional to density and stiffness. The strain mismatch maps to Γ_{mech}:

$$\Delta Z_{\text{bone}} = -\eta\,\Gamma_{\text{mech}}\,F_{\text{applied}}\,\varepsilon_{\text{bone}} \quad \checkmark \tag{42}$$

Astronaut bone loss: $F_{\text{applied}} \approx 0 \Rightarrow x_{\text{in}} \cdot x_{\text{out}} \approx 0 \Rightarrow \Delta Z = 0$. Without the C2 drive, bone impedance drifts under metabolic noise at $\sim 1\%$/month [40].

3. *Vascular: Glagov's Remodeling (1987)*

Arteries enlarge compensatorily as plaque accumulates, maintaining lumen until burden exceeds $\sim$ 40% [41]. Vascular impedance $Z_{\text{vasc}} = 8\mu L/\pi r^4$ (Poiseuille). Wall shear stress τ_w is x_{in}; endothelial response is x_{out}:

$$\Delta Z_{\text{vasc}} = -\eta\,\Gamma_{\text{vasc}}\,\tau_w\,(\text{endothelial response}) \quad \checkmark \tag{43}$$

The 40% threshold corresponds to the critical Γ beyond which the smooth muscle learning rate η cannot keep pace with plaque growth.

4. *Immune: Affinity Maturation*

Receptor–antigen impedance: $Z_{\text{receptor}} = 1/(K_a \cdot [\text{R}])$. Affinity maturation via somatic hypermutation [42]: $\Delta K_a \propto K_a \cdot [\text{Ag}] \cdot [\text{T}_{\text{FH}}]$. Since $\Delta Z \propto -\Delta K_a$:

$$\Delta Z_{\text{receptor}} = -\eta\,\Gamma_{\text{immune}}\,[\text{Ag}]\,[\text{T}_{\text{FH}}] \quad \checkmark \tag{44}$$

TABLE II. Interface–relay map for the 29 Γ-networks.

#	Network	Interface	Relay structure
1	Neural	Neuron–neuron	Synapse
2	Skeletal	Bone–soft tissue	Periosteum
3	Vascular	Blood–vessel wall	Endothelium
4	Immune	Immune–tissue	Lymph node
5	Epithelial	Body–environment	Basement membrane
6	Muscular	Muscle–tendon	Myotendinous junction
7	Pulmonary	Blood–air	Alveolar membrane
8	GI	Lumen–blood	Villous epithelium
9	Endocrine	Gland–blood	Fenestrated capillary
10	Lymphatic	Tissue–lymph	Lymphatic valve
11	Renal	Blood–filtrate	Glomerulus
12	Hepatic	Portal blood–bile	Disse space
13	Connective	Muscle–bone	Tendon sheath
14	Adipose	Fat–vasculature	Lobular septum
15	Haematopoietic	Marrow–blood	Sinusoidal endothelium
16	Reproductive	Follicle–blood	Theca–granulosa junction
17	Cardiac	Electrical–mechanical	Purkinje fibre
18	Reg. brain	Set-point–error	Hypothalamic nuclei
19	Ocular	Aqueous–venous	Trabecular meshwork
20	Auditory	Sound–neural	Organ of Corti
21	Vestibular	Motion–neural	Semicircular canal crista
22	Olfactory	Gas–neural	Olfactory epithelium
23	Dental	Oral–pulp	Odontoblast layer
24	Enteric	Lumen–ENS	Myenteric plexus
25	Autonomic	CNS–viscera	Autonomic ganglia
26	Pancreatic	Glucose–insulin	Islet of Langerhans
27	CSF	Brain–venous	Choroid plexus
28	Brain (func.)	Cortical areas	Association fibres
29	Cartilage	Load–matrix	Perichondrium ($\eta \approx 0$)

A vaccine is the deliberate injection of x_{in} to drive C2 in the absence of disease. Booster doses are spaced repetition for Γ reduction [43].

5. *Cartilage: the C2 null case ($\eta \approx 0$)*

Articular cartilage is avascular; chondrocytes are sparse and post-mitotic. Consequently $\eta_{\text{cart}} \approx 0$ [44]. When loading changes ($\Gamma_{\text{cart}} > 0$), $\Delta Z \approx 0$ regardless—the tissue cannot adapt. Osteoarthritis is not a "disease" but the *predicted consequence* of an interface with $\eta \approx 0$.

a. The healthy joint as impedance transformer. An articular joint couples bone ($Z = 120$) to synovial fluid ($Z = 50$) through a cartilage layer at $Z_{\text{cart}} = 80 \approx \sqrt{120 \times 50} = 77.5$—the geometric mean condition for a quarter-wave transformer [45]. When cartilage erodes (end-stage OA), the relay node is removed: $\Gamma = (120 - 50)/(120 + 50) = 0.41$, T drops from > 0.80 to 0.59, producing chronic reflection-driven pain.

XVII. CENTRAL CONTROLLERS

The preceding tissue networks describe *distributed* C2—local impedance matching along branches. Two systems operate at a higher hierarchical level, setting Z_{source} or Z_{setpoint} for entire network trees.

A. Cardiac pump: Laplace–Starling remodeling

The heart is the voltage source whose output impedance determines the pressure waveform for the entire arterial tree. C2 mapping: $\Delta Z_{\text{heart}} = -\eta\,\Gamma_{\text{aortic}}\,\sigma\,\varepsilon$, where σ is wall stress (Laplace: $\sigma = Pr/2h$) and ε is myocardial strain.

Two hypertrophy modes:

- **Concentric** (pressure overload, $\Gamma > 0$): wall thickness h increases, Z_{heart} rises toward Z_{aorta}.
- **Eccentric** (volume overload): chamber dilates, reducing mismatch from a different direction.

Heart failure is $\Gamma_{\text{aortic}} \to 1$: ventricular impedance can no longer match the arterial tree. Controller failure propagates to all downstream vascular branches.

B. Regulatory brain: baroreceptor resetting and allostatic load

The brainstem sets Z_{setpoint} for the entire autonomic tree via the baroreflex arc. C2 mapping: $\Delta Z_{\text{setpoint}} = -\eta\,\Gamma_{\text{reg}}$ (sensory input) (autonomic output).

Baroreceptor resetting in chronic hypertension [46]: the brainstem adapts its pressure setpoint upward ($Z_{\text{setpoint}} \uparrow$), reducing local Γ_{reg} but creating downstream mismatch in every organ.

Allostatic load [20], mathematically the *cumulative exogenous set-point forcing*:

$$L_{\text{allostatic}} = \int_0^T |\Delta Z_{\text{setpoint}}(t)|\,dt\,, \tag{45}$$

i.e. the time-integrated drift of the regulatory controller's impedance set-point. Chronic stress causes multi-organ disease because the *controller's* Γ error propagates to all downstream networks.

Remark 12 (The full chain: trauma → multi-organ disease)**.** *Part B (Sec. XIV C) derives a caregiver lifecycle equation with an explicit empathic-load term $+\lambda\,\Gamma^2_{other}$. When this term is chronically elevated (unresolved PTSD, long-term caregiving), the brainstem controller accumulates allostatic load, drifting $Z_{setpoint}$ upward and raising Γ_v across every downstream organ. Because immune memory and metabolic homeostasis are themselves C2 networks (Table I), the resulting $\rho \downarrow$ starves their repair loops: immune affinity*

maturation stalls ($\eta_{immune}\downarrow$), insulin and hepatic regulation lose their corrective gradient, and erythropoiesis underperforms. The outward signs—lowered immunity, metabolic syndrome, accelerated skin and hair aging—are not metaphors for "stress"; they are quantitative consequences of $H = (1-\Gamma_n^2)(1-\Gamma_v^2)\downarrow$ propagating from the controller through every C2 branch of the network.

XVIII. DISCUSSION

A. Methodological caveat

All computational experiments (Paper 5) use modules built from the Γ-Net axioms. When the simulators reproduce the theory's predictions, this confirms code–equation consistency, not empirical truth. True validation requires longitudinal impedance measurements (EEG coherence from birth to senescence), comparative polysomnography across phyla, and prospective clinical tests.

B. The brain as thermal manager

The Γ-framework identifies the brain as a centralised thermal manager for impedance-debt dissipation, necessitated by the loss of the oceanic heat sink. Cognition, memory, and consciousness exploit infrastructure whose *physical* raison d'être is thermodynamic housekeeping [13]. This is consistent with the preoptic area controlling *both* sleep and thermoregulation: thermal management is the brain's primary physical function; higher cognition operates on the same impedance substrate.

C. $\Gamma^2 \leftrightarrow T$: the self-consistent loop

The relationship between impedance mismatch and body temperature is not a one-way dependency; it forms a closed, self-consistent loop. We now make this explicit.

a. Step 1: Computation → heat. C1 (Paper 0, Eq. 1) guarantees that every interface with $\Gamma_i \neq 0$ dissipates $\Gamma_i^2 P_{\text{in},i}$ as heat. Summed over N active interfaces:

$$P_{\text{heat}} = \sum_{i=1}^{N} \Gamma_i^2 \, P_{\text{in},i} = \langle \Gamma^2 \rangle \cdot P_{\text{total}} \,. \tag{46}$$

b. Step 2: Heat → temperature. In thermal steady state:

$$T_{\text{body}} = T_{\text{env}} + \frac{P_{\text{heat}}}{h_c A} = T_{\text{env}} + \frac{\langle \Gamma^2 \rangle \, P_{\text{total}}}{h_c A} \,, \tag{47}$$

where $h_c A$ is the effective thermal conductance (primarily cerebral blood flow, Paper 2 [2]).

c. Step 3: Temperature → impedance. Blood viscosity sets the vascular bus impedance $Z_{\text{bus}}(T)$ via the Arrhenius relation, and C2 repair rate scales as $\tilde{\eta}_{\text{learn}}(T) = \exp[\beta(T_0^{-1} - T^{-1})]$. Therefore:

$$\Gamma_i(T) = \frac{Z_{\text{organ},i} - Z_{\text{bus}}(T)}{Z_{\text{organ},i} + Z_{\text{bus}}(T)} \,. \tag{48}$$

d. Closure: the self-consistency condition. Substituting Eq. (48) into Eq. (46) and Eq. (47) yields a fixed-point equation:

$$\boxed{T^* = T_{\text{env}} + \frac{P_{\text{total}}}{h_c A} \sum_{i=1}^{N} \Gamma_i^2(T^*) \,.} \tag{49}$$

The operating point $T^* \approx 310\,\text{K}$ (37 °C) is the unique stable fixed point of this equation for terrestrial mammals. It is *not* a set-point chosen by evolution—it is the self-consistent solution where impedance-driven heat production balances convective heat loss.

e. Stability and failure modes. The loop is *stable* because $\partial P_{\text{heat}}/\partial T < h_c A$ at T^*: a small temperature rise shifts Z_{bus}, reducing some Γ_i^2 (better vascular matching at lower viscosity), which reduces heat production. The loop *fails* in two directions:

- **Hypothermia** ($T \ll T^*$): viscosity rises → $\Gamma^2 \uparrow$ globally → C2 repair slows (Arrhenius) → impedance debt accumulates → positive feedback → coma.
- **Hyperthermia** ($T \gg T^*$): thermal noise $\propto k_B T$ overwhelms high-K channels (Paper 4 [4]) → $\delta\Gamma/\Gamma \propto K^{3/2}$ → PFC collapses first → delirium.

The 37 °C operating point is the narrow basin of attraction between these two catastrophes.

D. Connection to Papers 1–2

Paper 1 established the *spatial* structure (relay nodes, fractal topology, dimensional cut cost, irreducibility). Paper 2 established the *vascular* extension (Murray's Law, dual-network organ health, positive-feedback cascade). This paper extends both in two directions: *time* (Parts A–B) and *breadth* (Part C). Part A derives daily dynamics (impedance debt, sleep, morphogenesis, evolution); Part B derives lifecycle dynamics (learning–novelty–aging competition, emotion, bathtub curve, death); Part C proves that all twenty-nine known tissue-remodeling laws are algebraically equivalent to C2 and identifies the two central controllers (cardiac pump, regulatory brain) whose impedance errors propagate to every downstream organ. Together: the static architecture (Paper 1) sets channel structure $\{Z_i\}$; the vascular model (Paper 2) provides material coupling $I_{\text{blood}} = (1-\Gamma_v^2)Q_0$; Parts A–B determine daily and lifecycle regulation; Part C completes the tissue catalogue. All layers derive from a single principle: $\mathcal{A}[\Gamma] = \int \Sigma|\Gamma|^2 \, dt \to \min$.

E. Unification of developmental psychology

The Lifecycle Equation reduces the entire developmental trajectory to three physical forces on one state variable. Piaget's stages are phase transitions of $\Sigma\Gamma^2$; Vygotsky's ZPD is the Γ_{env} range where learning can reduce $\Sigma\Gamma^2$ in one session; critical periods are intervals of enhanced $\eta(t)$.

F. Emotion as physics, not computation

Dopamine, cortisol, curiosity, and boredom are computed from Γ^2 time series in the Alice system. The cognitive–emotional distinction dissolves [26]: both are readouts of the same impedance dynamics at different time scales.

G. Clinical implications

Depression: high $\Sigma\Gamma_n^2$, low $|d\Gamma_n^2/dt|$ (treatment: increase η_n or decrease Γ_{env}). PTSD: undamped Γ^2 spike (treatment: titrated re-exposure). ADHD: excess γ (stimulants increase effective η). Dementia: accelerated $\delta_n D_n$ (often preceded by vascular cascade). Hypertension: chronic $\Sigma\Gamma_v^2$ elevation. Metabolic syndrome: dual-network failure (Γ_n^2 and Γ_v^2 both elevated, $H \to 0$).

H. No-waste corollary

Every molecular species possesses nonzero Z; its presence modifies Γ at every interface. "Metabolic waste" is an observational artefact arising from incomplete knowledge of the full Γ-network —consistent with the historical reclassification of nitric oxide, lactate, bilirubin, uric acid, and adenosine.

I. Dual-field morphogenesis

The unification of Wolff's law, Davis's law, and impedance remodeling as parameter instances of a single PDE (Table III) suggests a common thermodynamic origin: $\mathcal{A}[\Gamma] = \int |\Gamma|^2\, dt \to \min$. The material bottleneck ($\rho \to 0 \Rightarrow$ only degradation survives) connects directly to vascular disease via Paper 2: $\Gamma_v\uparrow$ arrests neural plasticity, structural remodeling, *and* homeostatic regulation simultaneously.

TABLE III. Four-tissue unification: same PDE, different parameters.

Tissue	D_ρ/D_Z	Γ_i^2	Γ_f^2	Reduction
Bone (Wolff)	500	0.0391	0.0340	13.1%
Neuron (C2)	2	0.0391	0.0281	28.2%
Muscle (Davis)	10	0.0391	0.0310	20.7%
Liver (hepatic)	200	0.0391	0.0195	50.1%

XIX. CONCLUSION

We have developed the temporal theory of impedance networks and proved the universality of the C2 remodeling law in three complementary parts.

Part A: Impedance debt $D_Z = \int |\Gamma|^2 P_{\text{in}}\, dt$ accumulates during wakefulness and is discharged by sleep. Its decomposition into D_{micro} and D_{thermal} is consistent with: cnidarian sleep without a brain (ocean removes D_{thermal}); the physical necessity of centralised neural architecture on land (loss of passive heat sink); the tidal zone as the lowest-action impedance transition ($\mathcal{A}[\Gamma]$ reduced 93.8%); and embryogenesis as thermodynamic causal chain. A dual-field PDE unifies four tissue-remodeling laws.

Part B: The Lifecycle Equation (Eq. (24)) reduces the entire cognitive trajectory to three forces: learning, novelty, aging. Emotion is impedance-mismatch readout at different time scales. The bathtub curve emerges without fitted functions. Aging is Coffin–Manson fatigue with Arrhenius acceleration; atherosclerosis is vascular fatigue; death is positive-feedback collapse. Adulthood has a physical definition via the Equilibrium Equation (27). PTSD, empathic load, and chronic stress are derived as instances of a single impedance-debt integral with different time-scale parameters.

Part C: The C2 remodeling rule $\Delta Z = -\eta\, \Gamma\, x_{\text{in}}\, x_{\text{out}}$ is algebraically equivalent to all twenty-nine known tissue-remodeling laws (Table I). Life is thermodynamically bounded between ice ($D_Z = 0$, C2 halted) and fire ($Z \to \infty$, substrate destroyed), with 37°C as the self-consistent fixed point (Eq. (49)). The cardiac pump and regulatory brain are central controllers whose impedance errors propagate to every downstream organ; allostatic load (Eq. (45)) is the time-integrated drift of the controller set-point. Cartilage ($\eta \approx 0$) predicts osteoarthritis as the null case of remodeling.

Together with Papers 1–2, this paper completes a multi-layer description of biological intelligence: spatial architecture, vascular topology, daily maintenance, the full arc of a life, and the universal tissue catalogue —all from $\mathcal{A}[\Gamma] \to \min$. Testable predictions and computational verification are consolidated in Paper 5 [3].

ACKNOWLEDGMENTS

The author thanks the open-source scientific Python community (NumPy, SciPy, pytest) whose tools made the Alice Γ-Net simulator possible.

DATA AND CODE AVAILABILITY

All simulation code, including the `SleepPhysicsEngine`, `ImpedanceDebtTracker`, `LifecycleEquationEngine`, `PinchFatigueEngine`, `CuriosityDriveEngine`, and `MetacognitionEngine` modules, is available at `https://github.com/cyhuang76/alice-gamma-net` under AGPL-3.0. Neonatal EEG data sourced from the PhysioNet PICS archive (open access).

[1] Hsi-Yu Huang, "Minimum Reflection Action on impedance networks: Irreducibility of dimensional cut cost and fractal scaling," Paper 1 of this series, Zenodo (2026).

[2] Hsi-Yu Huang, "Dual impedance networks: A unified neural–vascular organ model from the Minimum Reflection Action," Paper 2 of this series, Zenodo (2026).

[3] Hsi-Yu Huang, "The grand unification: Γ as the universal interface law from microtubules to power grids, and C2 as universal remodeling in twenty-nine tissues," Paper 5 of this series, Zenodo (2026).

[4] Hsi-Yu Huang, "Topological pathology: Disease cascades, coupled failure, and recovery on Γ-networks," Paper 4 of this series, Zenodo (2026).

[5] L. Appelbaum *et al.*, "Sleep is driven by neuronal DNA damage and repair in jellyfish and sea anemones," *Nature Communications* (2026).

[6] R. D. Nath *et al.*, "The jellyfish *Cassiopea* exhibits a sleep-like state," *Current Biology* **27**, 2984 (2017).

[7] L. L. Moroz, "Phylogenomics meets neuroscience: How many times might complex brains have evolved?" *Acta Biologica Hungarica* **63**, 3 (2012).

[8] J. Karbowski, "Thermodynamic constraints on neural dimensions, firing rates, brain temperature and size," *J. Comput. Neurosci.* **27**, 415 (2009).

[9] T. Porkka-Heiskanen and A. V. Kalinchuk, "Adenosine, energy metabolism and sleep homeostasis," *Sleep Medicine Reviews* **15**, 123 (2011).

[10] A. A. Borbély, "A two process model of sleep regulation," *Human Neurobiology* **1**, 195–204 (1982).

[11] J. Wolff, *Das Gesetz der Transformation der Knochen* (Hirschwald, Berlin, 1892).

[12] A. M. Turing, "The chemical basis of morphogenesis," *Phil. Trans. R. Soc. Lond. B* **237**, 37–72 (1952).

[13] J. S. Torday and W. B. Miller, "On the evolution of the mammalian brain," *Frontiers in Systems Neuroscience* **10**, 31 (2016).

[14] L. F. Coffin, "A study of the effects of cyclic thermal stresses on a ductile metal," *Trans. ASME* **76**, 931 (1954).

[15] W. Schultz, P. Dayan, and P. R. Montague, "A neural substrate of prediction and reward," *Science* **275**, 1593 (1997).

[16] R. M. Yerkes and J. D. Dodson, "The relation of strength of stimulus to rapidity of habit-formation," *J. Comp. Neurol. Psychol.* **18**, 459 (1908).

[17] D. Kahneman, *Thinking, Fast and Slow* (Farrar, Straus and Giroux, 2011).

[18] G. A. Klutke, P. C. Kiessler, and M. A. Wortman, "A critical look at the bathtub curve," *IEEE Trans. Reliab.* **52**, 125 (2003).

[19] R. Kail, "Developmental change in speed of processing during childhood and adolescence," *Psychol. Bull.* **109**, 490 (1991).

[20] B. S. McEwen, "Stress, adaptation, and disease: Allostasis and allostatic load," *Ann. N.Y. Acad. Sci.* **840**, 33 (1998).

[21] T. A. Salthouse, *Major Issues in Cognitive Aging* (Oxford University Press, 2010).

[22] P. R. Huttenlocher, "Synaptic density in human frontal cortex—developmental changes and effects of aging," *Brain Res.* **163**, 195 (1979).

[23] J. Piaget, *The Origins of Intelligence in Children* (International Universities Press, 1952).

[24] L. S. Vygotsky, *Mind in Society: The Development of Higher Psychological Processes* (Harvard University Press, 1978).

[25] T. K. Hensch, "Critical period plasticity in local cortical circuits," *Nat. Rev. Neurosci.* **6**, 877 (2005).

[26] A. R. Damasio, *Descartes' Error: Emotion, Reason, and the Human Brain* (Avon Books, 1994).

[27] M. P. Walker, *Why We Sleep: Unlocking the Power of Sleep and Dreams* (Scribner, 2017).

[28] M. M. Ohayon *et al.*, "Meta-analysis of quantitative sleep parameters from childhood to old age," *Sleep* **27**, 1255 (2004).

[29] B. A. Mander, J. R. Winer, and M. P. Walker, "Sleep and human aging," *Neuron* **94**, 19 (2017).

[30] E. S. Epel *et al.*, "Accelerated telomere shortening in response to life stress," *Proc. Natl. Acad. Sci. U.S.A.* **101**, 17312 (2004).

[31] SPRINT MIND Investigators, "Effect of intensive vs standard blood pressure control on probable dementia," *JAMA* **321**, 553 (2019).

[32] S. J. Lupien *et al.*, "Effects of stress throughout the lifespan on the brain, behaviour and cognition," *Nat. Rev. Neurosci.* **10**, 434 (2009).

[33] A. L. Alter and D. M. Oppenheimer, "Uniting the tribes of fluency to form a metacognitive nation," *Pers. Soc. Psychol. Rev.* **13**, 219 (2009).

[34] C. R. Harris *et al.*, "Array programming with NumPy," *Nature* **585**, 357 (2020).

[35] J. F. Sorce *et al.*, "Maternal emotional signaling: its effect on the visual cliff behavior of 1-year-olds," *Dev. Psychol.* **21**, 195 (1985).

[36] D. O. Hebb, *The Organization of Behavior* (Wiley, New York, 1949).

[37] L. Xie *et al.*, "Sleep drives metabolite clearance from the adult brain," *Science* **342**, 373 (2013).

[38] N. E. Fultz *et al.*, "Coupled electrophysiological, hemodynamic, and cerebrospinal fluid oscillations in human sleep," *Science* **366**, 628 (2019).

[39] M. Nedergaard, "Garbage truck of the brain," *Science* **340**, 1529 (2013).

[40] T. Lang *et al.*, "Cortical and trabecular bone mineral loss from the spine and hip in long-duration spaceflight," *J. Bone Miner. Res.* **19**, 1006 (2004).

[41] S. Glagov *et al.*, "Compensatory enlargement of human atherosclerotic coronary arteries," *N. Engl. J. Med.* **316**, 1371 (1987).

[42] G. D. Victora and M. C. Nussenzweig, "Germinal centers," *Annu. Rev. Immunol.* **30**, 429 (2012).

[43] N. J. Cepeda *et al.*, "Distributed practice in verbal recall tasks," *Rev. Educ. Res.* **76**, 354 (2006).

[44] J. A. Buckwalter and H. J. Mankin, "Articular cartilage: tissue design and chondrocyte–matrix interactions," *Instr. Course Lect.* **47**, 477 (1998).

[45] D. M. Pozar, *Microwave Engineering*, 4th ed. (Wiley, 2011).

[46] M. W. Chapleau, J. T. Cunningham, M. J. Sullivan, R. E. Wachtel, and F. M. Abboud, "Structural versus functional modulation of the baroreflex," *Hypertension* **18**, 667 (1991).

[47] A. Krogh, "The number and distribution of capillaries in muscles with calculations of the oxygen pressure head necessary for supplying the tissue," *J. Physiol.* **52**, 409–415 (1919).